

Shear wave propagation physics and numerical implementations in the quasi-incompressible acousto-elastic soft tissue regime

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Abstract

Ultrasound-driven shear wave propagation in soft tissue underlie shear wave elastography (SWE) and emerging elastomechanical hypotheses of ultrasonic neuromodulation. Accurate modeling of radiation-force-induced tissue motion is therefore of central importance. However, general-purpose finite element elastodynamic solvers are often computationally expensive and unnecessarily broad for shear-dominant ultrasound applications. In soft biological tissue, the large separation between longitudinal and shear wave speeds ($\sim 10^3$ ratio) creates distinct temporal scales, enabling sequential acoustic-to-elastic modeling strategies. We derive a reduced viscoelastic elastodynamic formulation tailored to ultrasound-driven problems by systematically applying near-incompressibility, small-strain linearization, and solenoidal force projection to the full Navier equations. The resulting Kelvin–Voigt shear wave equation retains only the transverse dynamics relevant to radiation-force-induced motion while incorporating shear viscosity for attenuation modeling. An explicit finite-difference time-domain (FDTD) implementation is developed, with second-order accuracy in space and time for the elastic component and stable treatment of viscous dissipation under classical CFL constraints. Validation studies demonstrate recovery of theoretical shear-wave speeds, convergence under grid refinement, and physically consistent attenuation behavior. The reduced framework provides a computationally efficient and physically transparent alternative to full multiphysics elasticity solvers, accommodates spatial heterogeneity in shear modulus, density, and viscosity, and facilitates integration with acoustic simulation tools and supporting both clinical SWE modeling and mechanistic investigation of ultrasound-induced neuromodulation. Its lightweight implementation and computational efficiency broaden applications to inverse problems, iterative methods, and machine learning data generation tasks.

1 Introduction

Ultrasound-driven mechanical phenomena in soft tissue play a central role in several biomedical applications, most prominently shear wave elastography (SWE) [54, 4, 25, 56] and ultrasonic neuromodulation [7, 30, 48]. In SWE, acoustic radiation force [36, 42, 38, 61] generates transient elastic displacements whose propagation encodes the underlying shear modulus of tissue. In ultrasonic neuromodulation, related elastomechanical interactions—often framed through elastodynamic or radiation-force-mediated hypotheses—have been proposed as potential mechanisms linking acoustic exposure to neural activity [7, 31, 67]. These application domains share a common requirement: accurate and computationally efficient modeling of elastic wave generation and propagation in soft solids under ultrasound excitation.

Extensive prior work has developed high-fidelity simulation frameworks for SWE based on general-purpose finite element methods (FEM) [28, 68], including widely used packages such as LS-DYNA and Abaqus [5, 10]. These tools can represent complex geometries, heterogeneous materials, nonlinear constitutive behavior, and boundary interactions, making them highly versatile for broad classes of elastodynamic problems. However, this generality comes at substantial

computational and algorithmic cost. For ultrasound-specific problems—where the primary interest lies in small, radiation-force-induced shear displacements in nearly incompressible soft tissue—the full generality of large-scale FEM is often unnecessary and can obscure the essential physics governing wave generation, propagation, and measurement.

A central challenge in ultrasound-driven elastodynamics is the strong separation of physical scales between acoustic and shear phenomena. Acoustic waves in soft tissue propagate with effective longitudinal speeds on the order of 10^3 m/s [58, 18], governed by the large bulk modulus, whereas shear waves travel at speeds typically near 1–5 m/s [53, 60]. This disparity produces temporal scales that differ by approximately three orders of magnitude, despite spatial domains that remain comparable. As a result, fully coupled multiphysics simulations that resolve both acoustic and elastic dynamics simultaneously are computationally inefficient and often unnecessary. Instead, practical modeling strategies commonly treat the acoustic field and the resulting radiation-force-driven elastic response on separate timescales, coupling them through intermediate source terms: the acoustic pressure or intensity is computed first (e.g., with fast nearfield and angular spectrum methods or full-wave codes), the radiation force is derived from it, and that force is then supplied as a body force to a shear wave solver [66]. Such acoustic–elastic coupling approaches, including those based on established acoustic simulators (e.g., FIELD II [29], Fullwave [47]), are well documented in the SWE literature [5, 4]. Full finite-element modeling of ARF-induced shear wave propagation in tissue-mimicking media is covered by established guidelines [43], which describe a sequential pipeline (compute 3-D ARF push, apply to an FEM, extract motion) and recommend commercial codes (e.g., LS-DYNA, Abaqus); that approach is general but computationally heavy.

Numerical simulation is widely used to validate and improve shear wave imaging techniques. Most simulations of shear waves generated by an applied acoustic radiation force rely on the finite element (FEM) method. FEM models have been developed for ARFI imaging [44], for the shear wave response to radiation force in breast [37] and liver [45], for viscoelastic propagation [33], for thermal effects in soft tissue [41], and for related applications such as lesion characterization [20], tracking and reflected-wave effects [40], and transversely isotropic media [52]. Alternatives to full-field FEM include the finite difference method [39, 32], the pseudospectral method [49], and the Green’s function approach [24, 5, 9, 13]. Green’s function methods apply to both elastic [9] and viscoelastic media [5] and are well suited when the response is needed at only a few observation points; full-field methods (FEM, FDTD, pseudospectral) instead compute the entire transient response over the domain. The choice between these strategies therefore depends on whether one needs the full field, on spatial heterogeneity, and on the type of source, i.e. radiation force versus prescribed mechanical excitation.

Among finite-difference approaches, dedicated 3-D FDTD simulations of shear waves for complex modulus imaging have been implemented [39] in the context of inverse problems. That work assumes incompressibility and pure shear (normal stress components set to zero), yielding six first-order velocity–stress equations in 3-D with Kelvin–Voigt viscoelasticity on a staggered grid with PML boundaries; the source is a vibrating needle chosen to minimize compressional energy. The present formulation, by contrast, uses a displacement-based update with an explicit solenoidal projection of the body force $\mathbf{f}_S$ via a scalar Poisson (CG) solve each step, so that arbitrary (e.g., radiation-force) sources can be applied without prescribing a needle geometry. Both are reduced shear-only forward solvers: the Orescanin et al. approach targets algebraic Helmholtz inversion (AHI) for complex modulus reconstruction from harmonic excitation, whereas the present solver emphasizes integration with acoustic simulators and Kelvin–Voigt attenuation in the shear equation.

Green’s function methods offer a different trade-off. GPU-accelerated implementations [66] provide fast evaluation of shear waves from a prescribed radiation force in elastic and viscoelastic media within the same sequential acousto-elastic workflow: the approach superposes contributions from each source point and is well suited to homogeneous media and sparse observation

points. The present FDTD formulation instead advances the full displacement field on a grid at each time step, so that the response is available everywhere at once and the same framework can accommodate spatial heterogeneity and more general boundary conditions. The solver described here aligns with sequential, shear-focused strategies—minimal reduced equation (solenoidal projection, Kelvin–Voigt shear only), explicit FDTD, and matrix-free conjugate-gradient projection for the body force—yielding a lightweight option for integration with acoustic simulators.

The objective of the present work is thus to derive and implement a reduced, purpose-built formulation of shear elastodynamics tailored specifically to ultrasound-driven problems in soft solids. Rather than pursuing full general-purpose elasticity, we seek to isolate the minimal physical model required to accurately reproduce radiation-force-induced shear displacements and their propagation, while reducing computational complexity. The formulation accommodates spatial heterogeneity in shear modulus, density, and viscosity. This reduction enables tight integration with acoustic simulation tools and supports efficient multiphysics workflows relevant to SWE, elastomechanical hypotheses of ultrasonic neuromodulation, and iterative methods for inverse problems.

The remainder of this manuscript develops the governing physics, numerical formulation, and validation of this reduced elastodynamic framework.

2 Reduced Elastodynamic Formulation

This section derives the governing equations of motion for ultrasound-driven shear dynamics in soft tissue, beginning from the full linear elastodynamic continuum description and systematically applying the physical reductions established below. The main assumptions are near-incompressibility of soft tissue ($K \gg \mu$), small-strain linearization, and timescale separation between acoustic and shear dynamics; they are invoked explicitly in Sec. 2.1 and the incompressible limit (Sec. 2.4).

2.1 Near-Incompressibility in Soft Tissue and Modulus Hierarchy

A key simplification exploited in ultrasound-driven elastodynamics is the extreme separation between volumetric (compressional) stiffness and shear stiffness in soft tissue [23, 18]. In isotropic linear elasticity [2, 27, 1], the longitudinal and shear wave speeds satisfy

$$c_L^2 = \frac{K + \frac{4}{3}\mu}{\rho}, \quad (1)$$

$$c_S^2 = \frac{\mu}{\rho}, \quad (2)$$

where ρ is density, K is bulk modulus, and μ is shear modulus. Using typical ultrasound values in soft tissue [18] ($c_L \approx 1540$ m/s and $\rho \approx 1000$ kg/m³) gives

$$K \approx \rho c_L^2 \approx 2.4 \times 10^9 \text{ Pa}, \quad (3)$$

where the correction $\frac{4}{3}\mu$ is negligible because $\mu \ll K$. Meanwhile, typical shear-wave speeds in elastography [53, 4] ($c_S \approx 1\text{--}5$ m/s) imply

$$\mu = \rho c_S^2 \approx 1 \times 10^3\text{--}2.5 \times 10^4 \text{ Pa}. \quad (4)$$

Thus, in ultrasound-relevant soft tissue,

$$\frac{K}{\mu} \sim 10^5\text{--}10^6, \quad (5)$$

which is a far stronger hierarchy than one would infer from moderate values of Poisson’s ratio alone.

For completeness, the elastic constants satisfy

$$E = 2\mu(1 + \nu), \quad \nu = \frac{3K - 2\mu}{2(3K + \mu)}, \quad K = \frac{E}{3(1 - 2\nu)}. \quad (6)$$

Under the hierarchy $K \gg \mu$, these reduce to the familiar incompressible limits

$$\nu \approx \frac{1}{2} - \frac{\mu}{6K}, \quad E \approx 3\mu, \quad (7)$$

so that the Young's modulus remains in the kPa range even though the bulk modulus is in the GPa range. Numerically, combining the representative values above yields $\nu = 0.499999$ (approximately), emphasizing that “near-incompressible” in the ultrasound sense corresponds to ν extremely close to 0.5 when K is inferred from c_L .

Interpretation for ultrasound-driven shear wave propagation. The hierarchy $K \gg \mu$ implies that compressional disturbances propagate rapidly and are energetically expensive in volumetric strain, whereas the radiation-force-induced response of interest in elastography [53, 60] (and potentially in elastomechanical hypotheses of neuromodulation [7, 48]) is dominated by slower shear motion governed primarily by μ . This motivates (i) shear-focused elastic models and (ii) sequential acousto-elastic workflows where the acoustic field is used to compute forcing, and the subsequent shear dynamics are simulated on a much slower time axis.

In ultrasound-driven problems, the external body force arises from acoustic radiation force. Because the acoustic field evolves on timescales three orders of magnitude faster than shear propagation (Sec. 2), the force distribution may be computed independently using an acoustic simulation and treated as a prescribed source term in Eq. (16). The resulting formulation constitutes a sequential acoustic-to-elastic coupling appropriate for shear wave elastography and related elastomechanical applications in ultrasonic neuromodulation.

2.2 Full linear viscoelastodynamics

Let $\mathbf{u}(\mathbf{x}, t)$ denote the displacement field in an isotropic, linearly viscoelastic continuum of density ρ . We adopt a Kelvin–Voigt constitutive law [11, 22], in which the Cauchy stress is decomposed into an elastic and a viscous contribution,

$$\boldsymbol{\sigma} = \lambda \operatorname{tr}(\boldsymbol{\epsilon})\mathbf{I} + 2\mu\boldsymbol{\epsilon} + 2\eta\dot{\boldsymbol{\epsilon}}, \quad (8)$$

where $\boldsymbol{\epsilon} = \frac{1}{2}(\nabla\mathbf{u} + (\nabla\mathbf{u})^T)$ is the small-strain tensor, $(\cdot)$ denotes a time derivative, λ and μ are the Lamé parameters, and η is the shear viscosity. Conservation of linear momentum with body force density $\mathbf{f}$ yields

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = \nabla \cdot \boldsymbol{\sigma} + \mathbf{f}. \quad (9)$$

Substituting Eq. (8) into Eq. (9) gives

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = (\lambda + \mu)\nabla(\nabla \cdot \mathbf{u}) + \mu\nabla^2 \mathbf{u} + \eta\nabla^2 \frac{\partial \mathbf{u}}{\partial t} + \mathbf{f}, \quad (10)$$

where the viscous term contributes a Laplacian of particle velocity.

2.3 Helmholtz decomposition of displacement

The displacement field may be decomposed into irrotational (longitudinal) and solenoidal (shear) components [2, 27],

$$\mathbf{u} = \nabla\phi + \nabla \times \boldsymbol{\Psi}, \quad (11)$$

where ϕ is a scalar potential and Ψ is a vector potential satisfying $\nabla \cdot \Psi = 0$.

Substituting into Eq. (10) yields decoupled wave equations for the longitudinal and shear components:

$$\frac{\partial^2 \phi}{\partial t^2} = c_L^2 \nabla^2 \phi + \frac{1}{\rho} f_L, \quad (12)$$

$$\frac{\partial^2 \Psi}{\partial t^2} = c_S^2 \nabla^2 \Psi + \frac{1}{\rho} \mathbf{f}_S, \quad (13)$$

where f_L and $\mathbf{f}_S$ denote the longitudinal and solenoidal projections of the body force.

2.4 Incompressible limit and reduced shear equation with viscosity

Given the extreme modulus hierarchy $K \gg \mu$ (Sec. 2.1), the incompressible limit is taken:

$$\nabla \cdot \mathbf{u} = 0, \quad (14)$$

which eliminates longitudinal dynamics and introduces a pressure field p as a Lagrange multiplier that enforces the constraint. In this limit, Eq. (10) becomes

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = -\nabla p + \mu \nabla^2 \mathbf{u} + \eta \nabla^2 \frac{\partial \mathbf{u}}{\partial t} + \mathbf{f}. \quad (15)$$

Taking the divergence of this momentum equation and enforcing Eq. (14) yields a Poisson equation for p that keeps the displacement divergence-free at each time. Projecting Eq. (15) onto the divergence-free subspace removes the pressure gradient and gives the reduced forced shear wave equation

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = \mu \nabla^2 \mathbf{u} + \eta \nabla^2 \frac{\partial \mathbf{u}}{\partial t} + \mathbf{f}_S, \quad (16)$$

where $\mathbf{f}_S$ is the solenoidal projection of the applied body force. Equation (16) governs shear wave propagation with wavespeed $c_S = \sqrt{\mu/\rho}$ and frequency-dependent attenuation set by η . Under the assumptions of linear elasticity, near incompressibility, small displacement, and timescale separation between acoustic and shear dynamics, this equation forms the physical basis for the numerical formulation of ultrasound-driven tissue motion.

3 Numerical Methods

3.1 FDTD Time-Domain Update (Explicit, Second Order)

Let $\mathbf{u}^n$ denote displacement at time $t_n = n\Delta t$. The implemented update [59] is

$$\mathbf{u}^{n+1} = 2\mathbf{u}^n - \mathbf{u}^{n-1} + \Delta t^2 \mathbf{a}^n, \quad (17)$$

with acceleration

$$\mathbf{a}^n = \frac{\mu}{\rho} \nabla_h^2 \mathbf{u}^n + \frac{\eta}{\rho} \nabla_h^2 \dot{\mathbf{u}}^n + \frac{1}{\rho} \mathbf{f}_S^n, \quad \dot{\mathbf{u}}^n \approx \frac{\mathbf{u}^n - \mathbf{u}^{n-1}}{\Delta t}. \quad (18)$$

∇_h^2 is a second-order finite-difference 3D Laplacian on cell centers.

For homogeneous media without viscosity, a standard stability bound is approximately

$$\Delta t \lesssim \frac{1}{c_s} \left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2} \right)^{-1/2}, \quad (19)$$

which motivates the CFL-based [16] step choice in validation scripts. The overall scheme is second-order accurate in time for the elastic component and first-order accurate in time for viscous dissipation, consistent with Kelvin–Voigt damping [11]. Numerical convergence and stability are verified empirically in Sec. 4 through grid-refinement and timestep studies.

3.2 CG Projection for the Scalar Poisson Problem

3.2.1 Helmholtz Projection of Body Force

At each step, the code projects $\mathbf{f}$ into solenoidal and irrotational parts:

$$\mathbf{f} = \mathbf{f}_S + \nabla\phi, \quad \nabla \cdot \mathbf{f}_S = 0. \quad (20)$$

Taking divergence gives Poisson equation:

$$\nabla^2\phi = \nabla \cdot \mathbf{f}, \quad (21)$$

then

$$\mathbf{f}_S = \mathbf{f} - \nabla\phi. \quad (22)$$

Numerically, this is solved by conjugate-gradient [26, 55] on the scalar Poisson problem.

Physical consequence. The solver removes forcing components that would excite compressional motion. It is a shear-mode propagator, not a full elastic-wave solver.

3.2.2 Why Conjugate Gradient on a Scalar Poisson Problem?

The projection step enforces $\nabla \cdot \mathbf{f}_S = 0$ by writing

$$\mathbf{f}_S = \mathbf{f} - \nabla\phi, \quad (23)$$

which leads to the scalar Poisson equation

$$\nabla^2\phi = \nabla \cdot \mathbf{f}. \quad (24)$$

Thus, only one scalar field ϕ must be solved each step (rather than a full vector system). After finite-difference discretization, this becomes a large sparse linear system

$$A\phi = b, \quad (25)$$

where A is the discrete negative Laplacian on interior nodes (with masking/boundary treatment).

Conjugate gradient (CG) [55, 26] is a natural choice because:

- A is symmetric positive-definite on the interior unknowns, matching CG requirements;
- the 3D problem is large and sparse, so direct factorization is often memory-expensive;
- CG only needs repeated stencil-like matrix-vector operations, which scale well in 3D.

From a physics perspective, this solve removes the irrotational (gradient) force component that would drive compressional response in full elastodynamics. The remaining divergence-free force preferentially excites transverse shear dynamics, consistent with the shear-focused SWE objective.

3.2.3 Numerical Method Details for the Scalar Poisson CG Solve

In implementation, the projection solve is performed in matrix-free form as

$$A\phi = b, \quad A = -\nabla_h^2, \quad b = \nabla_h \cdot \mathbf{f}, \quad (26)$$

where ∇_h^2 is a second-order central-difference 3D Laplacian (7-point stencil on interior nodes).

Masked interior update. The code uses an interior mask m (1 in interior, 0 on outer layer) and applies it after each update:

$$\phi \leftarrow \phi \odot m, \quad r \leftarrow r \odot m, \quad p \leftarrow p \odot m. \quad (27)$$

This enforces the chosen boundary treatment and avoids boundary unknown drift.

CG iteration (matrix-free). Starting from $\phi_0 = 0$:

$$r_0 = (b - A\phi_0) \odot m, \quad p_0 = r_0, \quad (28)$$

$$\alpha_k = \frac{r_k^T r_k}{p_k^T A p_k}, \quad (29)$$

$$\phi_{k+1} = (\phi_k + \alpha_k p_k) \odot m, \quad (30)$$

$$r_{k+1} = (r_k - \alpha_k A p_k) \odot m, \quad (31)$$

$$\beta_k = \frac{r_{k+1}^T r_{k+1}}{r_k^T r_k}, \quad (32)$$

$$p_{k+1} = (r_{k+1} + \beta_k p_k) \odot m. \quad (33)$$

No sparse matrix is assembled explicitly; Aq is computed by applying the Laplacian stencil to q .

Stopping and numerical safeguards. The solver terminates when relative residual drops below tolerance,

$$\sqrt{\frac{\|r_k\|_2^2}{\|r_0\|_2^2}} < \text{tol}, \quad (34)$$

or when maximum iterations are reached. Additional guards stop iterations if denominator or residual becomes non-finite or non-positive in a way that indicates numerical breakdown.

Precision and cost. The Poisson CG routine is evaluated in float64 for improved robustness, then cast back to the field dtype. Computational cost is approximately $O(N n_{\text{iter}})$ with N grid points, which is efficient for repeated projection steps in 3D without the memory burden of direct factorization.

3.3 Pre-computed Helmholtz Projection for Separable Body Forces

The CG projection described in Sec. 3.2 is applied at every time step in the general solver, which is correct for arbitrary time-varying body forces $\mathbf{f}(\mathbf{x}, t)$. However, in the practically important case of acoustic radiation force (ARF) driven shear wave excitation, the spatial and temporal dependences of the body force are separable. The ARF spatial distribution $\mathbf{b}_0(\mathbf{x})$ is determined entirely by the acoustic intensity field and tissue absorption, while the temporal dependence is controlled by the push envelope (e.g., a gated window during which the acoustic source is active). The body force then takes the form

$$\mathbf{f}(\mathbf{x}, t) = \mathbf{b}_0(\mathbf{x}) g(t), \quad (35)$$

where $g(t)$ is a scalar temporal envelope (for example, a step function that equals one during the push and zero afterwards).

3.3.1 Linearity of the Helmholtz Projection

The Helmholtz–Hodge decomposition is a linear operation: for any vector field $\mathbf{f}$ and scalar α , the solenoidal projection $\mathcal{P}_S$ satisfies

$$\mathcal{P}_S[\alpha \mathbf{f}] = \alpha \mathcal{P}_S[\mathbf{f}]. \quad (36)$$

This follows directly from the linearity of the Poisson equation that defines the projection. Given the decomposition $\mathbf{f} = \mathbf{f}_S + \nabla\phi$ with $\nabla^2\phi = \nabla \cdot \mathbf{f}$, scaling the input by α scales both ϕ and $\mathbf{f}_S$ by the same factor.

Applying this to the separable force in Eq. (35), the solenoidal projection at any time t_n is

$$\mathbf{f}_S^n = \mathcal{P}_S[\mathbf{b}_0 g(t_n)] = g(t_n) \mathcal{P}_S[\mathbf{b}_0] = g(t_n) \mathbf{f}_{S,0}, \quad (37)$$

where $\mathbf{f}_{S,0} \equiv \mathcal{P}_S[\mathbf{b}_0]$ is the divergence-free projection of the static spatial pattern, computed once.

3.3.2 Algorithm

The pre-computed projection replaces the per-step CG solve with a single projection followed by scalar multiplication at each time step:

1. **Pre-computation (once).** Solve the scalar Poisson problem

$$\nabla_h^2 \phi_0 = \nabla_h \cdot \mathbf{b}_0, \quad (38)$$

via conjugate gradient (as in Sec. 3.2), then form

$$\mathbf{f}_{S,0} = \mathbf{b}_0 - \nabla_h \phi_0. \quad (39)$$

2. **Time stepping.** At each step n , the solenoidal force is obtained by scaling:

$$\mathbf{f}_S^n = g(t_n) \mathbf{f}_{S,0}. \quad (40)$$

The FDTD update (Sec. 3.1) then proceeds with the acceleration

$$\mathbf{a}^n = \frac{\mu}{\rho} \nabla_h^2 \mathbf{u}^n + \frac{\eta}{\rho} \nabla_h^2 \dot{\mathbf{u}}^n + \frac{g(t_n)}{\rho} \mathbf{f}_{S,0}. \quad (41)$$

3.3.3 Computational Cost

For a grid with N points and N_t time steps, the per-step projection requires N_t CG solves at cost $O(N \cdot n_{\text{iter}})$ each, giving a total projection cost of $O(N \cdot n_{\text{iter}} \cdot N_t)$. The pre-computed approach requires a single CG solve at cost $O(N \cdot n_{\text{iter}})$, plus $O(N)$ scalar–vector multiplication per step, reducing the total projection cost to $O(N \cdot n_{\text{iter}} + N \cdot N_t)$.

In the ARF-driven regime the CG projection typically dominates wall-clock time (the Laplacian stencil evaluations in the FDTD update are comparatively cheap). For practical problem sizes—for example, $N = 106^3 \approx 1.2 \times 10^6$ points and $N_t = 85$ steps—the pre-computed projection reduces total simulation time from $O(10^2\text{--}10^3)$ s to $O(10)$ s, an improvement of one to two orders of magnitude.

3.3.4 Conditions for Validity

The pre-computed projection is exact whenever the body force is temporally separable in the sense of Eq. (35). This condition is satisfied in the standard ARF scenario where:

- the acoustic intensity field is computed once (steady-state or time-averaged over acoustic cycles),
- the resulting radiation force spatial pattern $\mathbf{b}_0(\mathbf{x})$ is fixed, and
- a scalar envelope $g(t)$ modulates the push on/off timing.

The approach is *not* applicable when the spatial force distribution itself changes with time, as would occur with a moving acoustic focus, time-varying aberration correction, or forces derived from a non-stationary pressure field. In such cases, the general per-step CG projection of Sec. 3.2 remains necessary.

3.4 Boundary and initial conditions

In the current implementation, the computational domain is terminated with a fixed (Dirichlet) boundary condition by setting displacement to zero on the outer layer. This choice simplifies the staggered-grid update and provides a well-posed finite domain problem, but it can introduce reflections that contaminate late-time arrivals. Absorbing boundary treatments such as perfectly matched layers (PML) [6, 15] can reduce such artifacts when late-time fidelity is required. Accordingly, validation and analysis are typically restricted to early and intermediate propagation times and/or to sensor offsets that are not dominated by boundary interactions. The initial state is commonly specified as a divergence-free velocity field constructed from a vector potential (a curl-based initialization), which preferentially excites shear motion while suppressing compressional content.

3.5 Validation approach and current implementation

Validation is treated as a layered process to separate implementation defects from modeling limitations. At the most basic level, code-correctness checks should include analytic or manufactured solutions in simplified geometries, symmetry checks (e.g., a symmetric source should produce a symmetric wavefield in a homogeneous medium), and regression tests that detect unintended changes. Numerical-consistency checks then assess stability and accuracy through grid and time-step refinement, sensitivity to the Courant–Friedrichs–Lewy (CFL) condition [16], and characterization of numerical dispersion and attenuation as resolution changes. Physics-consistency checks verify that homogeneous-media propagation recovers the expected shear wavespeed $c_s = \sqrt{\mu/\rho}$, that attenuation trends behave monotonically with the viscosity parameter η when present, and that passive cases do not exhibit non-physical energy growth. Finally, boundary and geometry checks quantify sensitivity to domain size and to the choice of boundary treatment (e.g., fixed versus absorbing layers).

The repository implements a focused wavespeed validation in `fullwave/shear_wave/validation/validate`. A homogeneous benchmark medium is defined by selecting a target shear speed c_s and setting $\mu = \rho c_s^2$. Shear-dominant excitation is obtained via curl-based initialization and a divergence-free force projection. Displacement traces are recorded at multiple offsets, and arrival times are estimated from a velocity proxy $\partial u_z / \partial t$ using peak picking with distance-based start gating. Recovered speed is reported per sensor, along with mean and linear-regression estimates [65, 14]. This validation establishes correct recovery of shear-wave propagation speed in controlled homogeneous cases [10], but does not by itself validate compressional physics, anisotropy, strong heterogeneity, or broadband viscoelastic constitutive behavior.

4 Results

4.1 Non-Viscous Validation Run ($\eta = 0$): Plots and Wavespeed Results

To isolate elastic wavespeed recovery, the large-field validation was run with $\eta = 0$ (no viscous term) on a cubic 180^3 grid ($L_x = L_y = L_z = 36$ mm, $\Delta x = 0.2$ mm). The generated plots are shown below.

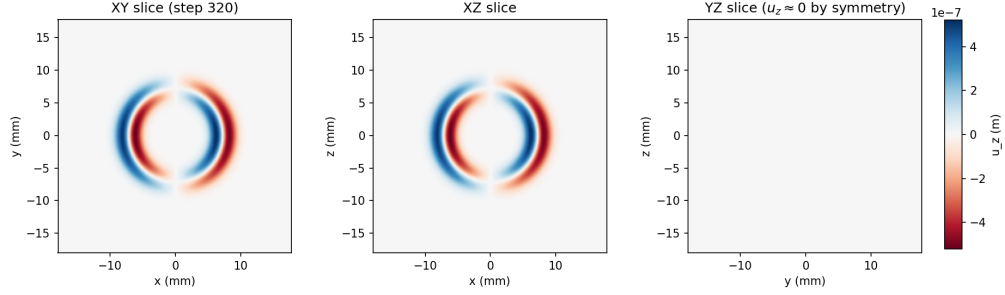


Figure 1: Representative shear-wave field snapshot (u_z , non-viscous run): XY slice (left), XZ slice (middle), YZ slice at $x = 0$ (right). The YZ slice is near zero because the curl-based vector-potential initialization drives motion in the x - z plane; the center x -plane is a nodal surface for u_z by symmetry.

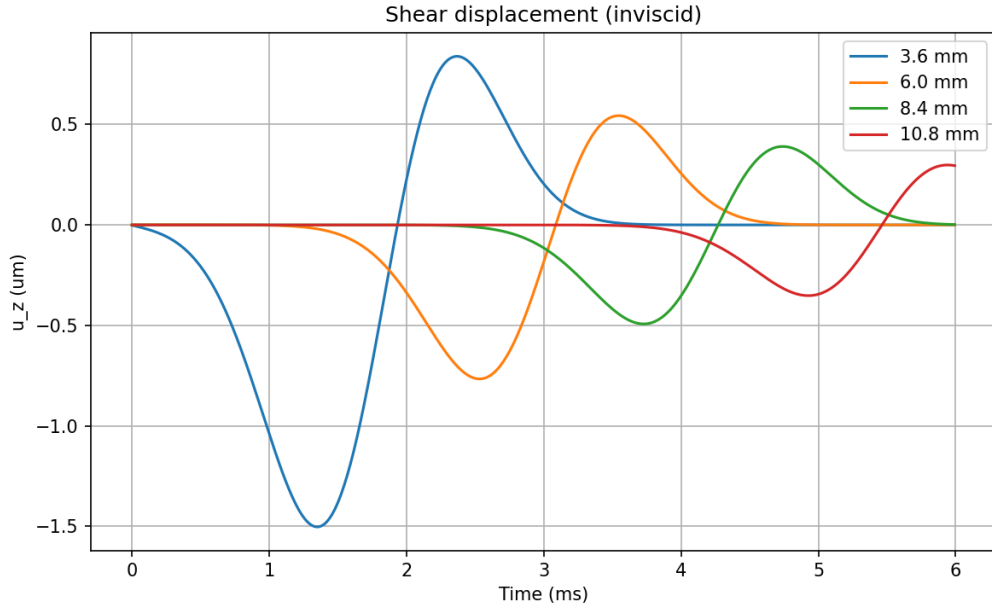


Figure 2: Displacement traces (u_z) at increasing sensor offsets for the non-viscous validation run.

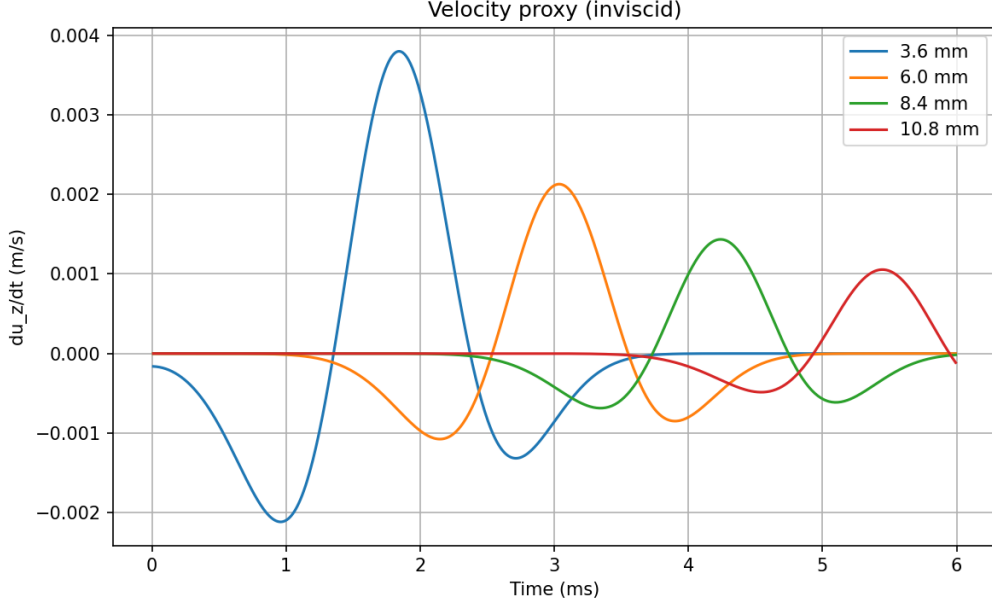


Figure 3: Velocity-proxy traces ($\partial u_z / \partial t$) used for arrival picking and wavespeed estimation.

Recovered values from the non-viscous run:

- Target speed: 2.000 m/s (from $\mu = \rho c_s^2$).
- Sensor-wise estimates (m/s): 1.961, 1.976, 1.982, 1.986, 2.300, 2.603.
- For validation metrics, the last two sensors are excluded (boundary/reflection sensitivity at long propagation distance).
- Mean speed using first four sensors only: 1.976 m/s.
- Linear fit speed using first four sensors only: 1.998 m/s.

Result interpretation. Using the first four sensors (nearest/intermediate range), the recovered speed is in close agreement with the imposed elastic target (2.000 m/s): the mean is 1.976 m/s ($\sim 1.2\%$ low) and the distance-time linear-fit speed is 1.998 m/s ($< 0.1\%$ error). This supports that the shear-wave propagation physics and numerics are correct in the non-viscous benchmark regime. The two farthest sensors are excluded here because they are more sensitive to late-time boundary/reflection effects in the current finite domain; if needed, this can be reduced further by enlarging the domain or using absorbing boundaries.

4.2 Grid Refinement / Convergence Study (Inviscid)

An inviscid convergence sweep was performed with fixed physical domain size ($L_x = L_y = L_z = 36$ mm), fixed physical sensor distances (first four sensors only), and a six-level refinement range. Crucially, the Gaussian source width ($\sigma_{xy} = 5$ grid points) and velocity damping coefficient ($\gamma_v = 0.001$) are held constant across all levels so that any differences are due purely to grid resolution.

- $n = 60$ ($dX = 6.00 \times 10^{-4}$ m): $\text{mean}_4 = 1.828$ m/s, $c_{\text{fit},4} = 2.193$ m/s.
- $n = 80$ ($dX = 4.50 \times 10^{-4}$ m): $\text{mean}_4 = 1.907$ m/s, $c_{\text{fit},4} = 2.048$ m/s.
- $n = 100$ ($dX = 3.60 \times 10^{-4}$ m): $\text{mean}_4 = 1.938$ m/s, $c_{\text{fit},4} = 2.053$ m/s.

- $n = 120$ ($dX = 3.00 \times 10^{-4}$ m): $\text{mean}_4 = 1.954$ m/s, $c_{\text{fit},4} = 2.017$ m/s.
- $n = 150$ ($dX = 2.40 \times 10^{-4}$ m): $\text{mean}_4 = 1.968$ m/s, $c_{\text{fit},4} = 2.007$ m/s.
- $n = 180$ ($dX = 2.00 \times 10^{-4}$ m): $\text{mean}_4 = 1.972$ m/s, $c_{\text{fit},4} = 1.998$ m/s.

Two complementary error metrics are used to quantify convergence:

1. **Wavespeed error** ($|\text{mean}_4 - c_{\text{target}}|$): the absolute difference between the mean recovered speed over the first four sensors and the analytical target. This is a scalar metric derived from discrete peak-picking, so it exhibits some variability.
2. **Trace L2 error**: the relative L_2 norm of the velocity trace difference at each sensor, averaged over the four sensors, measured against the finest grid ($n = 180$) as reference. This is a continuous metric that captures the full waveform convergence.

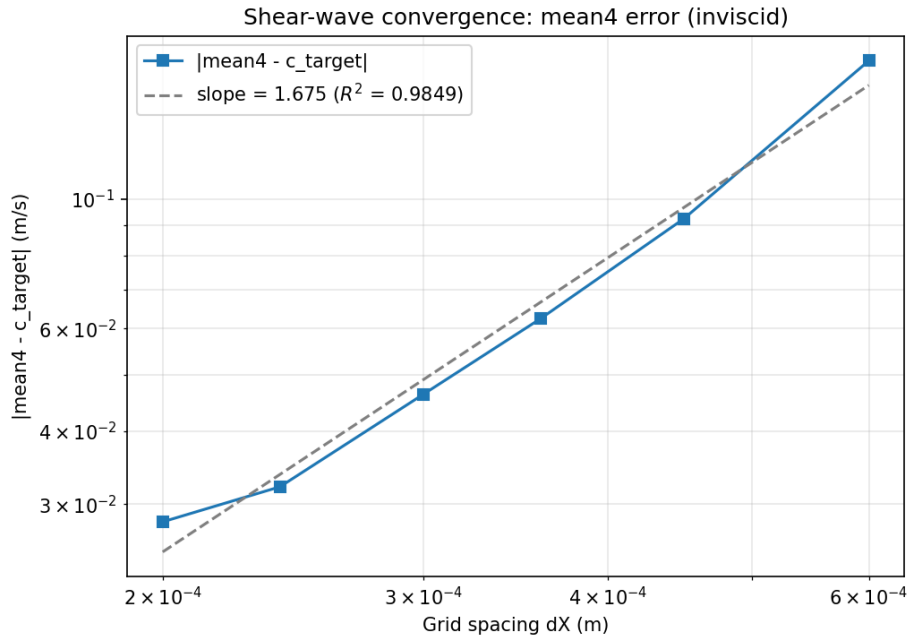


Figure 4: Wavespeed convergence: $|\text{mean}_4 - c_{\text{target}}|$ vs grid spacing (log-log). Fitted slope 1.68, $R^2 = 0.98$.

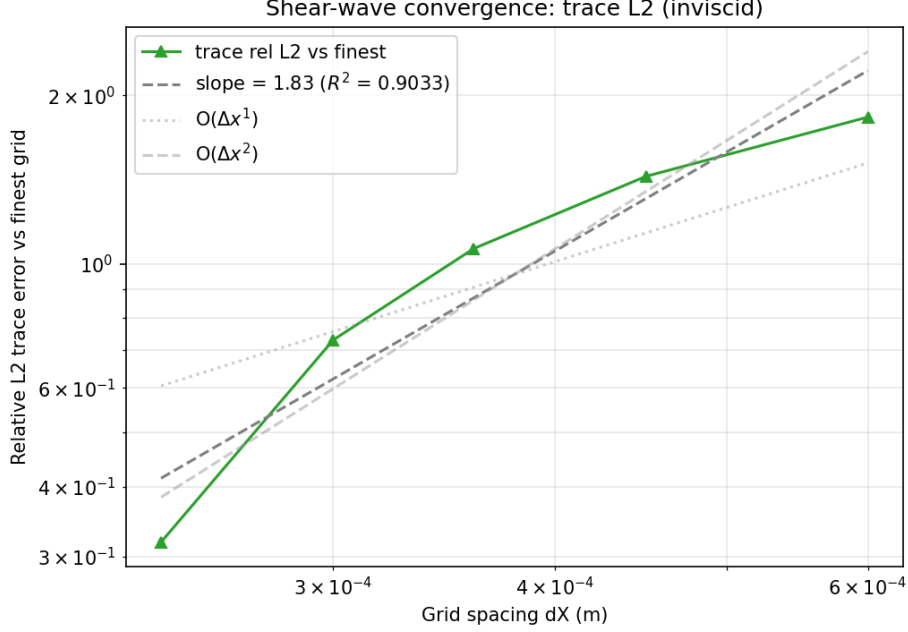


Figure 5: Trace L2 convergence: relative L_2 error of velocity traces vs finest grid (log-log). Fitted slope 1.83, $R^2 = 0.90$. Dashed reference lines show $O(\Delta x)$ and $O(\Delta x^2)$.

Quantitative convergence metrics. Fitting $|\text{mean}_4 - c_{\text{target}}| \propto dX^p$ in log-log space gives a global slope of $p = 1.68$ ($R^2 = 0.98$), with pairwise observed orders: $p_{60 \rightarrow 80} = 2.17$, $p_{80 \rightarrow 100} = 1.77$, $p_{100 \rightarrow 120} = 1.64$, $p_{120 \rightarrow 150} = 1.63$, $p_{150 \rightarrow 180} = 0.76$. The trace L2 metric gives a global slope of $p = 1.83$ ($R^2 = 0.90$), monotonically decreasing from 1.83 at $n = 60$ to 0.32 at $n = 150$.

Interpretation. Both metrics confirm convergence consistent with the second-order spatial discretization (7-point Laplacian stencil) and second-order leapfrog time integration. The trace L2 metric, which captures the full waveform rather than a single scalar, yields a cleaner slope ($p \approx 1.83$) closer to the theoretical $O(\Delta x^2)$ rate. The wavespeed metric ($p \approx 1.68$) shows slightly sub-quadratic behavior due to the discrete nature of peak-picking for arrival time estimation. At the finest level ($n = 180$), the linear-fit speed reaches 1.998 m/s ($< 0.1\%$ error).

4.3 Viscosity Validation: 1D Harmonic Kelvin–Voigt Benchmark

To avoid known bias from finite 3D pulse-domain effects, viscosity validation is reported here using only the dedicated 1D harmonic benchmark (`fullwave/shear_wave/validation/validate_kelvin_voigt`), which better matches Kelvin–Voigt analytic assumptions (single frequency, effectively plane-wave propagation, homogeneous medium).

4.3.1 1D harmonic benchmark results

To align with Kelvin–Voigt theory and cover a brain-relevant range, the benchmark was run on $\eta \in [0, 1.5]$ Pa·s with $\Delta\eta = 0.1$ Pa·s. The high- η configuration uses (i) adaptive sensor placement based on the expected e-folding length, (ii) adaptive drive amplitude scaling, and (iii) a smooth frequency schedule $f_0(\eta)$ guided by dimensional analysis.

Define the loss parameter and dimensionless attenuation

$$\text{De}_v = \frac{\omega\eta}{\mu}, \quad \Pi_\alpha = \frac{\alpha c_s}{\omega}, \quad c_s = \sqrt{\mu/\rho}. \quad (42)$$

For weak-to-moderate loss ($\text{De}_v \ll 1$),

$$\Pi_\alpha \approx \frac{\text{De}_v}{2} \Rightarrow \alpha \approx \frac{\eta \omega^2}{2\rho c_s^3}. \quad (43)$$

Thus, to keep attenuation growth smooth with viscosity while staying tied to the Kelvin–Voigt asymptotic, we set a target power law

$$\alpha_{\text{target}}(\eta) \propto \eta^p, \quad (44)$$

which gives

$$f_0(\eta) \propto \eta^{-\beta}, \quad \beta = \frac{1-p}{2}. \quad (45)$$

In the implementation (`f0_mode=dimensional`), p is set by `f0_alpha_exp` (default $p = 0.6$, hence $\beta = 0.2$) unless overridden by `f0_dim_beta`. The provisional $f_0(\eta)$ is then corrected by solving the exact Kelvin–Voigt dispersion relation so that exact $\alpha(f_0, \eta)$ matches the asymptotic target. For $\eta_{\text{max}} = 1.5$ Pa s with 800 Hz base drive and `f0_min_scale=0.4`, this yields a smooth drop to the 742 Hz range.

Adaptive sensor placement. A key challenge for attenuation measurement at high viscosity is that the e-folding length $\ell_e = 1/\alpha$ shrinks rapidly: at $\eta = 1.5$ Pa s the theoretical $\alpha \approx 824$ Np/m gives $\ell_e \approx 1.2$ mm. Sensors placed beyond a few e-folding lengths measure only noise, biasing the attenuation fit low.

To address this, sensor positions are adapted per η based on the theoretical attenuation at the scheduled drive frequency. For each case, $\alpha_{\text{theory}}(f_0, \eta)$ is computed from the exact Kelvin–Voigt dispersion relation, and n sensors are placed linearly from $\frac{1}{2}\ell_e$ to $3\ell_e$, ensuring all sensors lie within a measurable amplitude range. For the inviscid case ($\eta = 0$), sensors fall back to a default spread covering 3%–18% of the domain.

Drive and frequency scheduling parameters. The run reported here used:

- Frequency scheduler: `f0_mode=dimensional`, $\eta_{\text{ref}} = 0.2$ Pa s, `f0_alpha_exp` = 0.6 $\Rightarrow \beta = 0.2$, smoothing factor 0.7, lower-bound scale `f0_min_scale` = 0.4.
- Drive scaling: $\eta_{\text{hi}} = 0.2$ Pa s, multiplier $1 + 6(\eta - \eta_{\text{hi}})$, capped at $6\times$.
- SNR diagnostics: $\text{SNR}_{\text{min}} = 6$ (used for sensor-quality flagging); attenuation figure shows the no-SNR-gate measured curve for visual continuity.

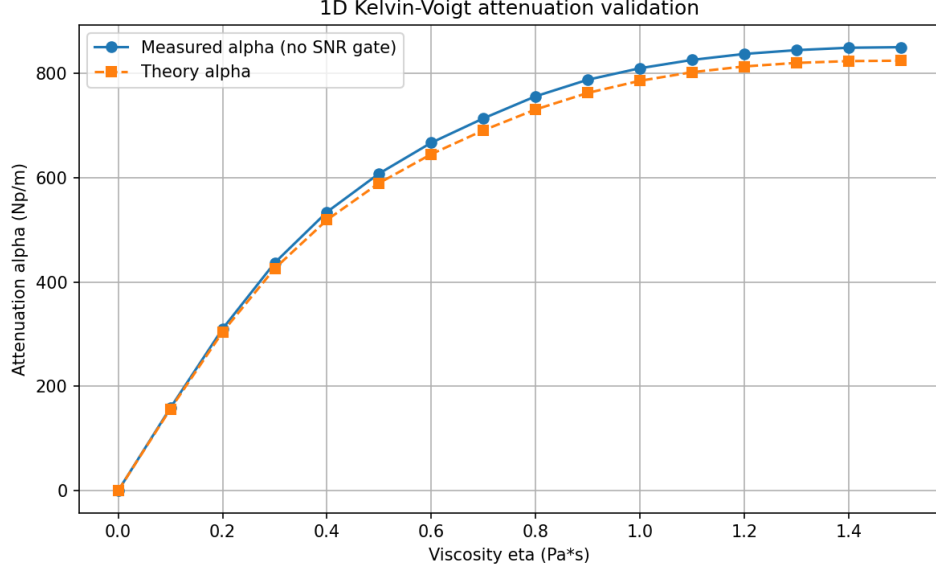


Figure 6: 1D harmonic Kelvin–Voigt benchmark: attenuation comparison (measured vs theory) over $\eta \in [0, 1.5]$ Pa s with adaptive sensor placement and frequency scheduling.

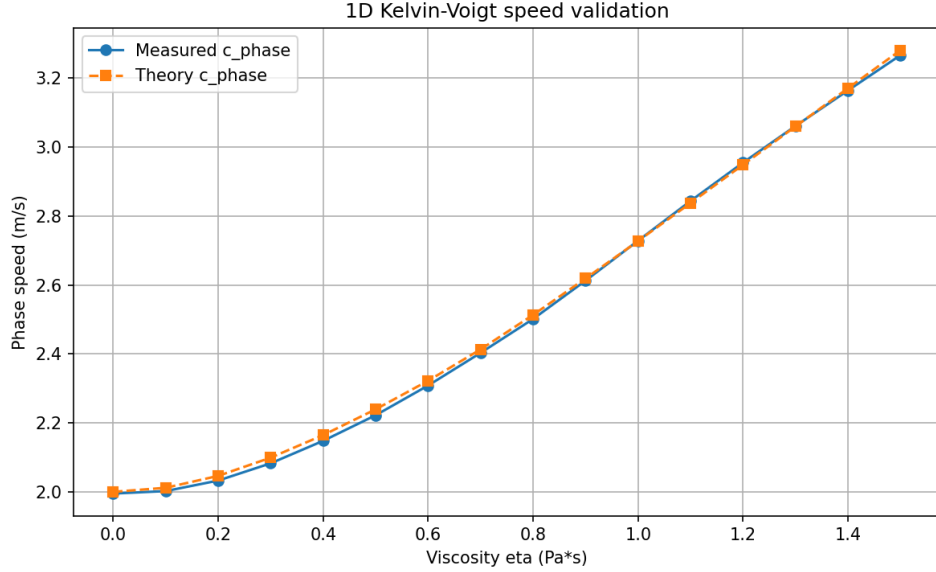


Figure 7: 1D harmonic Kelvin–Voigt benchmark: phase-speed comparison (measured vs theory).

Representative results from the 16-point viscosity sweep:

- $\eta = 0.10$ Pa s: $\alpha_{\text{meas}} = 158.5$ Np/m vs $\alpha_{\text{th}} = 156.4$ Np/m (+1.3%); $c_{\text{meas}} = 2.002$ m/s vs $c_{\text{th}} = 2.012$ m/s (−0.5%).
- $\eta = 0.30$ Pa s: $\alpha_{\text{meas}} = 436.8$ Np/m vs $\alpha_{\text{th}} = 425.8$ Np/m (+2.6%); $c_{\text{meas}} = 2.083$ m/s vs $c_{\text{th}} = 2.099$ m/s (−0.8%).
- $\eta = 0.50$ Pa s: $\alpha_{\text{meas}} = 607.8$ Np/m vs $\alpha_{\text{th}} = 589.5$ Np/m (+3.1%); $c_{\text{meas}} = 2.222$ m/s vs $c_{\text{th}} = 2.239$ m/s (−0.8%).
- $\eta = 1.00$ Pa s: $\alpha_{\text{meas}} = 809.7$ Np/m vs $\alpha_{\text{th}} = 785.6$ Np/m (+3.1%); $c_{\text{meas}} = 2.729$ m/s vs $c_{\text{th}} = 2.728$ m/s (< 0.1%).

- $\eta = 1.50$ Pa s: $\alpha_{\text{meas}} = 850.0$ Np/m vs $\alpha_{\text{th}} = 824.2$ Np/m (+3.1%); $c_{\text{meas}} = 3.267$ m/s vs $c_{\text{th}} = 3.281$ m/s (−0.4%).
- Monotonicity: both α_{meas} and α_{th} are monotone increasing over the full range $\eta \in [0, 1.5]$ Pa s.
- Aggregate agreement over $\eta \in [0.1, 1.5]$ Pa s: mean absolute attenuation error $\sim 3\%$, mean absolute phase-speed error $< 0.5\%$.

Conclusion. The combination of adaptive sensor placement (keyed to the theoretical e-folding length) and dimensional-analysis-guided frequency scheduling (Equation (43)) yields robust attenuation recovery across the full viscosity range. Both attenuation and phase speed track the analytical Kelvin–Voigt predictions with consistent $\sim 3\%$ and $< 0.5\%$ agreement, respectively, confirming the implicit viscous time-stepping scheme.

5 Discussion

5.1 Key physical assumptions

The solver is derived under a small-strain linearization: displacement gradients are assumed to be small and the stress–strain relationship is linear. This is typically appropriate for ultrasound radiation-force excitations in SWE [54, 4], where observed tissue motion is commonly in the micrometer to tens-of-micrometers regime, but it can fail under unusually strong pushes, large strains, near-contact compression, or when tissue damage and nonlinear mechanics become relevant.

Material behavior is modeled as isotropic with an effectively homogeneous (or piecewise homogeneous) shear response, represented by a scalar shear modulus $\mu(\mathbf{x})$ and density $\rho(\mathbf{x})$ [23, 18]. This approximation is often reasonable in parenchymal soft tissues at the spatial resolution used for SWE processing [25], but it can break down in anisotropic or strongly heterogeneous structures such as skeletal muscle, tendon, white-matter tracts, fibrotic septa, layered media, and near rigid interfaces (e.g., skull or other hard boundaries).

A central modeling choice is to represent shear-dominant propagation while omitting explicit compressional (longitudinal) elastodynamics. This is motivated by the near-incompressibility of soft tissue ($K \gg \mu$), which yields a strong separation between longitudinal and shear wave speeds ($c_L \gg c_S$) and hence their relevant timescales. In many SWE workflows, fast compressional transients exit the analysis window quickly or are reduced by directional and temporal filtering, whereas the quantity of interest is the slower transverse shear wave used for wavespeed and modulus estimation. The tradeoff is that this reduced model does not capture near-source pressure-wave content, P–S mode conversion at interfaces or strong heterogeneities, nonlinear shear or shear shock dynamics [63, 19, 64], or any diagnostic relying on volumetric strain.

Viscous attenuation is represented using a Kelvin–Voigt-like shear viscosity [11, 22], in which damping enters through a Laplacian term $\eta \nabla^2 \dot{\mathbf{u}}$. This single-parameter form is useful for capturing qualitative attenuation and pulse broadening, but soft tissue often exhibits frequency-dependent, power-law, or fractional viscoelastic behavior [65]; consequently, a constant η may be insufficient for broadband inversions or dispersion-sensitive analyses.

Finally, the present implementation enforces fixed (zero-displacement) boundaries. This simplifies numerics but can generate reflections that bias late-time arrival picking. Practical mitigation strategies include using larger computational domains, restricting analysis to earlier times/shorter propagation distances, or implementing absorbing layers (e.g., PML) [6, 15] when late-time fidelity is required.

5.2 Connection to ultrasound-driven shear rheology and SWE

In ARF-based SWE [36, 5], the acoustic field produces a predominantly axial body force in the push region. The resulting mechanical response contains both (i) a rapid local compression-related transient near the source and (ii) slower transverse shear-wave radiation away from the push region. When the scientific or clinical objective is shear-wave arrival timing and wavespeed in soft tissue, the reduced governing equation used here provides an efficient approximation that isolates the dominant shear dynamics. Conversely, when early-time fields, compressional contamination, mode conversion, poroelastic fluid–solid coupling, or complex boundary interactions are central to the question, a full elastodynamic or poroelastic model is required.

5.3 When the assumptions are likely valid, and when they are likely invalid

The reduced model is most appropriate in regimes with small displacements and approximately linear response [54]; in nearly incompressible soft tissue where SWE biomarkers depend primarily on the shear-wave speed [53, 60, 4]; in moderate propagation ranges where boundary reflections are not dominant; and in analyses that are effectively narrowband such that a Kelvin–Voigt damping term provides an adequate attenuation approximation [11].

The assumptions are more likely to fail in strongly anisotropic or heterogeneous media (e.g., muscle fibers, scar bands, skull-coupled configurations), in problems where compressional waves must be represented explicitly (especially near-source early-time dynamics), in regimes where nonlinear shear or shear shock formation is significant [63, 19], in broadband rheological inversions requiring more faithful frequency-dependent constitutive models, in domains where fixed-wall reflections strongly influence arrival times, and in settings where poroelasticity or other multiphase physics is needed to represent fluid–solid relative motion.

5.4 Practical interpretation of inputs and outputs

The solver outputs displacement and velocity fields that should be interpreted as shear-dominant wavefields. In homogeneous media, the characteristic shear-wave speed is $c_S \approx \sqrt{\mu/\rho}$, with practical bias introduced by viscosity, heterogeneity, finite-domain effects, and the specific arrival-picking strategy. In ultrasound multiphysics workflows, the body-force input should be understood as a radiation-force–equivalent source term [42, 51] derived from an acoustic simulation [29, 62, 47] (or from a prescribed analytic approximation), applied on the elastic timescale after the acoustic field has effectively traversed the domain. This separation of scales supports tight integration with acoustic tools while maintaining computational efficiency for shear dynamics.

5.5 Typical acoustic radiation force magnitudes

The body force that drives the shear solver is the acoustic radiation force (ARF), given by the plane-wave approximation [38, 61]

$$b = \frac{2 \alpha_{\text{Np}}}{c} I_{\text{SPPA}}, \quad I_{\text{SPPA}} = \frac{\langle p^2 \rangle}{\rho c}, \quad (46)$$

where α_{Np} is the absorption in Np/m at the transmit frequency f_0 , and I_{SPPA} is the spatial-peak pulse-average intensity at the focus. Because absorption in soft tissue scales approximately linearly with frequency [58, 18], the ARF magnitude depends strongly on f_0 even at the same focal intensity. Table 1 illustrates this effect for a representative tissue absorption of $0.5 \text{ dB cm}^{-1} \text{ MHz}^{-1}$ and a focal I_{SPPA} of 1000 W/cm^2 .

Table 1: Frequency dependence of ARF body force at fixed focal intensity ($I_{\text{SPPA}} = 1000 \text{ W/cm}^2$, $\alpha_0 = 0.5 \text{ dB cm}^{-1} \text{ MHz}^{-1}$, $c = 1540 \text{ m/s}$).

f_0 (MHz)	α_{NP} (Np/m)	b (N/m ³)
1	5.8	750
3	17.3	2 250
5	28.8	37 400
6.7	38.6	50 100

In clinical soft-tissue ARFI at 5–7 MHz, focal intensities of 500–2000 W/cm² and correspondingly high attenuation produce body forces on the order of 10^4 – 10^5 N/m^3 , yielding tissue displacements of 1–20 μm [36, 44, 42].

For *transcranial* applications at lower frequencies (0.5–1 MHz), the situation differs in two important ways. First, the absorption per unit path length is roughly an order of magnitude lower than at diagnostic frequencies, so the ARF per unit intensity is proportionally weaker. Second, the skull attenuates and aberrates the transmitted beam, reducing the focal intensity to roughly 30–50% of its free-field value [48]. The net effect is that body forces at 1 MHz are typically 10^2 – 10^4 N/m^3 (two orders of magnitude below high-frequency SWE), and the resulting brain tissue displacements are sub-micrometer to a few micrometers. This is consistent with transcranial MR-ARFI measurements reporting 0.5–1.2 μm displacements in primate brain at comparable acoustic parameters.

The transcranial demonstration in Appendix A provides a concrete example: a 1024-element sparse array at 1 MHz with 1.5 MPa source pressure produces a peak focal pressure of $\sim 5 \text{ MPa}$, a focal I_{SPPA} of $\sim 633 \text{ W/cm}^2$ (after CW envelope correction), and a peak ARF of $\sim 41,800 \text{ N/m}^3$ after skull propagation, yielding $\sim 1.7 \mu\text{m}$ displacement in brain tissue ($\mu \approx 4.6 \text{ kPa}$, $\eta = 0.5 \text{ Pa}\cdot\text{s}$).

5.6 From tissue displacement to membrane tension

In neuromodulation applications, the quantity that ultimately gates mechanosensitive ion channels is not tissue displacement itself but the *membrane tension* T (force per unit length, mN/m) experienced by the lipid bilayer in which the channel is embedded. Bridging the macroscopic displacement field produced by this solver to membrane tension requires two steps: (i) computing the tissue-level strain tensor from the displacement gradient, and (ii) converting the deviatoric strain to an effective bilayer tension via the membrane area-expansion modulus.

The linearized strain tensor $\varepsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$ is computed from the solver displacement field $\mathbf{u}(\mathbf{x}, t)$. Because the tissue is nearly incompressible, the volumetric (dilatational) strain is small and the mechanically relevant component is the deviatoric strain norm ε_{dev} . The effective membrane tension is then

$$T = K_A \alpha \varepsilon_{\text{dev}}, \quad (47)$$

where K_A is the area-expansion modulus of the lipid bilayer (typically 0.1–0.5 N/m for biological membranes [50]) and α is a dimensionless coupling factor relating far-field tissue strain to local membrane area change.

The tension T enters the Boltzmann gating model for mechanosensitive channels. Each channel opens with a steady-state probability $P_o(T) = [1 + \exp(-\Delta A (T - T_{1/2})/k_B T_{\text{abs}})]^{-1}$, where ΔA is the effective gate area change and $T_{1/2}$ is the half-activation tension—the tension at which the channel is open 50% of the time.

$T_{1/2}$ is determined experimentally. Two methods are used in practice. The most common is *patch-clamp electrophysiology with applied suction*: a glass micropipette is sealed onto a membrane patch, negative pressure (mmHg) is applied, and channel currents are recorded as a function of pressure. Converting the applied pressure P to tension requires the Laplace relation

$T = Pr/2$, where r is the radius of curvature of the membrane dome inside the pipette. Because r depends on pipette geometry and seal quality and is rarely measured directly, this conversion introduces uncertainty of order $\pm 1\text{--}2$ mN/m [35]. The more direct approach uses *reconstituted lipid bilayers*, in which purified channel proteins are inserted into artificial membranes whose tension is controlled and measured directly, eliminating the Laplace ambiguity [57, 8].

For the channels most relevant to ultrasonic neuromodulation, the best-characterized half-activation tensions are: Piezo1, $T_{1/2} = 2.7 \pm 0.4$ mN/m, measured by direct tension clamp with simultaneous patch imaging [35] and confirmed in cell-free bilayers [57]; and TRAAK, $T_{1/2} \approx 5$ mN/m, from reconstituted bilayers [8]. Values for TREK-1 (~ 3.5 mN/m) and TREK-2 (~ 4 mN/m) carry larger uncertainty because they are derived from patch-clamp pressure data with assumed pipette geometry [46, 3].

These tension scales set the target for the simulation pipeline: the shear wave solver must resolve displacement gradients (and hence strains) accurately enough that the resulting membrane tensions can be compared with the channel $T_{1/2}$ values. For a representative $K_A = 0.25$ N/m and $\alpha = 1$, a deviatoric strain of $\sim 10^{-5}$ produces a membrane tension of ~ 2.5 μ N/m—well below channel thresholds—while strains of order 10^{-2} reach the few-mN/m range where channel gating becomes significant. This underscores the importance of accurate strain computation from the displacement field and motivates the use of the deviatoric invariant (which isolates shape-changing deformation) rather than the full strain tensor.

6 Conclusion

This manuscript derived the reduced shear elastodynamic equations implemented in `fullwave/shear_wave/shear` and documented the physical assumptions under which the formulation is appropriate for ultrasound-driven mechanics, including SWE [25, 56, 42] and related elastomechanical hypotheses in neuromodulation [7, 30, 48]. The resulting solver targets shear-dominant motion in nearly incompressible soft solids, enabling computationally efficient simulation of radiation-force-induced displacements while remaining compatible with sequential acoustic-elastic coupling strategies that reflect the strong separation of longitudinal and shear timescales.

Validation in homogeneous benchmark media demonstrates accurate recovery of the imposed shear wavespeed under controlled conditions, and convergence sweeps support numerical consistency over practical grid resolutions. The principal limitations are those implied by the model reduction and boundary treatment: compressional dynamics, mode conversion at interfaces, anisotropy, strong heterogeneity, broadband frequency-dependent viscoelasticity, and late-time reflection artifacts are not fully represented. Within its intended scope, the reduced solver provides a lightweight, purpose-built tool for multiphysics ultrasound workflows where shear wave generation, propagation, and measurement are the dominant quantities of interest [54, 5, 4, 36, 60].

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A End-to-End Transcranial Shear Wave Demo

This appendix presents an end-to-end demonstration of the complete acoustic–elastic pipeline applied to transcranial focused ultrasound through a realistic human skull. The simulation uses a 1024-element sparse phased array focused through an *ex vivo* micro-CT skull (Halle temporal bone specimen) to generate acoustic radiation force (ARF) at a 50 mm focal depth, which then drives shear wave displacement in brain tissue via the FDTD solver derived in this document.

The pipeline proceeds in four stages:

1. Sparse array acoustic transmit through micro-CT skull (GPU FDTD),
2. Computation of the acoustic radiation force from the pressure field,
3. Shear wave FDTD simulation of viscoelastic brain tissue displacement,
4. Visualization of all intermediate and final fields.

The corresponding script is `examples/wave_3d/sparse_transcranial_shear_3d.py`.

A.1 Sparse Array Geometry

The transducer is a 1024-element sparse phased array with $750\ \mu\text{m}$ element pitch. Element positions are loaded from an HDF5 connector file and mapped onto the simulation grid (66×66 mm lateral–elevational aperture). Per-element focusing delays steer the beam to a 50 mm depth focal point. Figure 8 shows the element layout in the transducer plane.

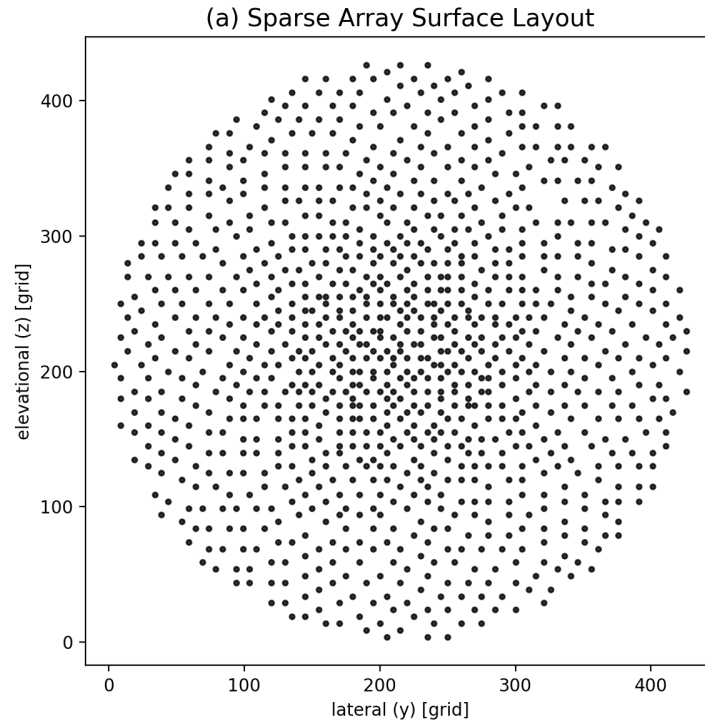


Figure 8: Sparse array surface layout showing element center positions in the lateral–elevational (y – z) plane.

A.2 Acoustical Medium Properties

The heterogeneous skull medium is constructed from a micro-CT NRRD volume of the Halle temporal bone specimen using the `MicroCTSkullDomain` preset. Hounsfield units are mapped to spatially varying sound speed (1540–2900 m/s), density (1000–2200 kg/m³), and attenuation coefficient. The background tissue is assigned standard soft-tissue properties. Figure 9 shows the resulting medium property maps.

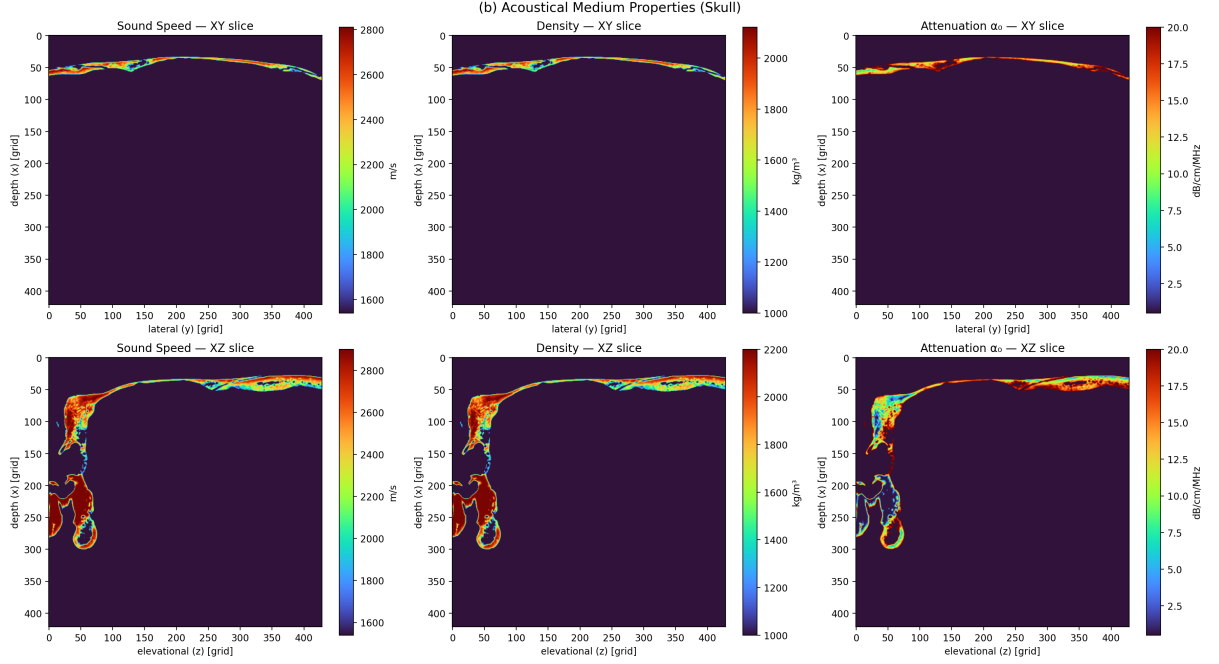


Figure 9: Acoustical medium properties derived from micro-CT skull data. **Top row:** x – y (depth–lateral) slices at center elevation showing sound speed, density, and attenuation coefficient α_0 . **Bottom row:** x – z (depth–elevational) slices at center lateral position. The skull layer is clearly visible as a high-speed, high-density region near the transducer surface.

A.3 Acoustic Pressure Propagation

The acoustic simulation uses the Fullwave GPU FDTD solver with exponential attenuation at $f_0 = 1$ MHz, source pressure amplitude of 1.5 MPa (typical clinical ARFI), and a sparse-grid sensor (4 $\times$ downsampling in each spatial dimension). Figure 10 shows the focused beam propagating through the skull into brain tissue.

(c) Pressure Propagation — XY Slices Through Skull

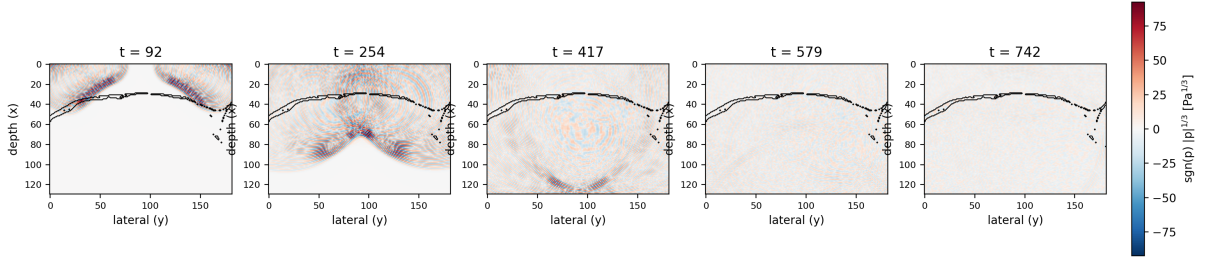


Figure 10: Pressure propagation snapshots in the x - y plane at center elevation. The focused beam emerges from the transducer (left edge), propagates through the skull (black contour), converges at the 50 mm focal zone, and subsequently diverges. Five time frames are shown spanning the full propagation duration.

A.4 Acoustic Radiation Force

Two formulations for the acoustic radiation force (ARF) are available.

Pressure-based (plane-wave) approximation. The simplest approach assumes locally planar wavefronts and computes a scalar body force along the beam axis:

$$b_0 = \frac{2\alpha_{\text{NP}}}{c} I_{\text{SPPA}}, \quad I_{\text{SPPA}} = \frac{1}{\rho c} \frac{\sum_{i=1}^{N_{\text{total}}} p_i^2}{N_{\text{pulse}}}, \quad (48)$$

where the sum runs over all N_{total} recorded time steps of the transient simulation and is normalized by the number of source-active steps N_{pulse} . This *silence correction* accounts for the dead time after the pulse has passed through the domain. A second *CW envelope correction* is needed because the Gaussian-windowed source pulse has a lower time-averaged $\langle p^2 \rangle$ than a continuous-wave signal at the same peak pressure. For CW transmission, $\langle p^2 \rangle = p_{\text{peak}}^2/2$, whereas for a short Gaussian-modulated pulse the mean is much smaller. The correction factor

$$C_{\text{CW}} = \frac{p_{\text{peak}}^2/2}{\langle p_{\text{src}}^2 \rangle} \quad (49)$$

scales the pulse-average intensity up to the equivalent CW value. For a 3-cycle pulse ($C_{\text{CW}} \approx 4.1$) and a 2-cycle pulse ($C_{\text{CW}} \approx 7.2$). The total normalization factor is the product of the silence and CW corrections. The same normalization is used for the volumetric heat source $Q = 2\alpha_{\text{NP}} I_{\text{SPPA}}$ in the bioheat module, ensuring a consistent mapping from transient simulations to equivalent continuous-wave intensity fields.

Velocity-based (Poynting vector) formulation. For converging beams the plane-wave assumption breaks down near the focal zone, where wavefronts are curved and energy flux has significant lateral components. A more accurate approach uses the time-averaged Poynting vector $\mathbf{I} = \langle p \mathbf{v} \rangle$, where $\mathbf{v}$ is the particle velocity field obtained directly from the FDTD solver (output_velocity=True). The vector body force is then

$$f_i = \frac{2\alpha_{\text{NP}}}{c} I_i, \quad I_i = \langle p v_i \rangle, \quad i \in \{x, y, z\}, \quad (50)$$

yielding three spatially varying force components (f_x, f_y, f_z).

In this demonstration we use the velocity-based formulation (`--use-velocity`) with the combined silence and CW envelope corrections (Eqs. 48–49). The radiation force within the skull is set to zero in both formulations (bone is rigid at the shear wave scale). Figure 11 shows the depth-component (f_x) of the resulting body force distribution.

Results. With the combined pulse normalization (silence $\times$ CW correction, total factor $\approx 57.8\times$ for this 2-cycle pulse), the velocity-based depth component gives $f_x^{\max} \approx 41,800 \text{ N/m}^3$, with lateral and elevational components $|f_y|^{\max} \approx 11,800 \text{ N/m}^3$ and $|f_z|^{\max} \approx 9,750 \text{ N/m}^3$. The off-axis components represent roughly 25% of the peak axial force and are passed to the shear wave solver as three-component driving terms.

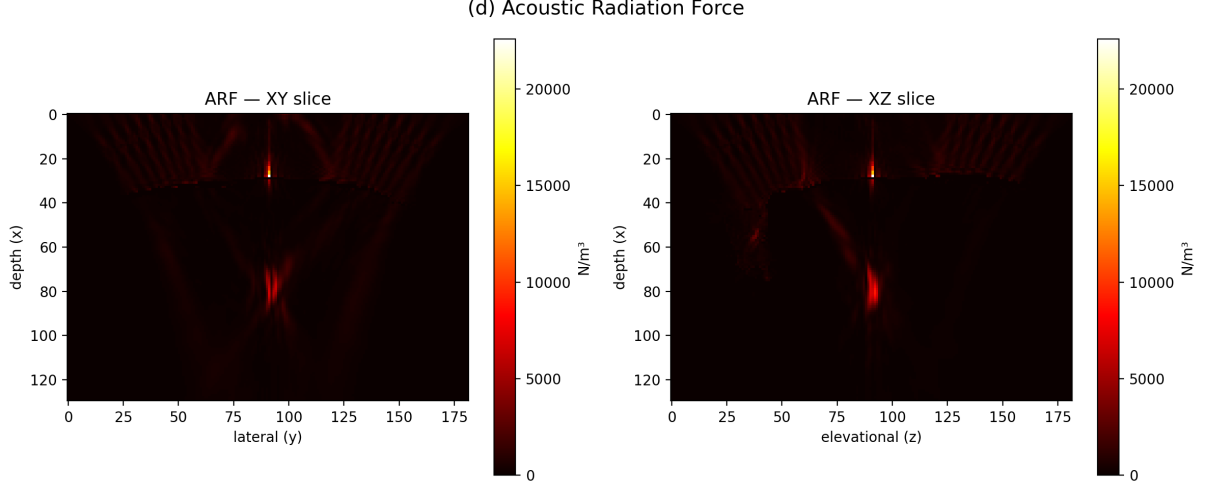


Figure 11: Acoustic radiation force (ARF) body force density (depth component f_x , velocity-based Poynting vector formulation with pulse-length and CW envelope normalization). **Left:** x - y slice. **Right:** x - z slice. The force is concentrated at the focal spot ($\sim 41,800 \text{ N/m}^3$ peak), with the skull region masked to zero.

A.5 Shear Wave Displacement

The shear wave simulation uses the staggered-grid FDTD solver (Section 3) with brain tissue parameters derived from the porcine brain measurements of Espíndola et al. [19] and Chandrasekaran et al. [12]: shear wave speed $c_s = 2.14 \text{ m/s}$ at 75 Hz, density $\rho = 1000 \text{ kg/m}^3$, giving shear modulus $\mu = \rho c_s^2 \approx 4580 \text{ Pa}$ and Young's modulus $E = 3\mu \approx 13.7 \text{ kPa}$ (assuming Poisson's ratio $\nu = 0.5$). Kelvin-Voigt viscosity is set to $\eta = 0.5 \text{ Pa}\cdot\text{s}$. All three ARF components (f_x, f_y, f_z) from the velocity-based formulation are applied as forcing terms via the separable temporal envelope (`env_t`), enabling a single pre-computed Helmholtz projection.

Figure 12 shows the shear wave displacement magnitude $|\mathbf{u}|$ at five time points, demonstrating the wave expanding radially from the focal zone.

(e) Shear Wave $|u|$ Snapshots — XY (top) / XZ (bottom)

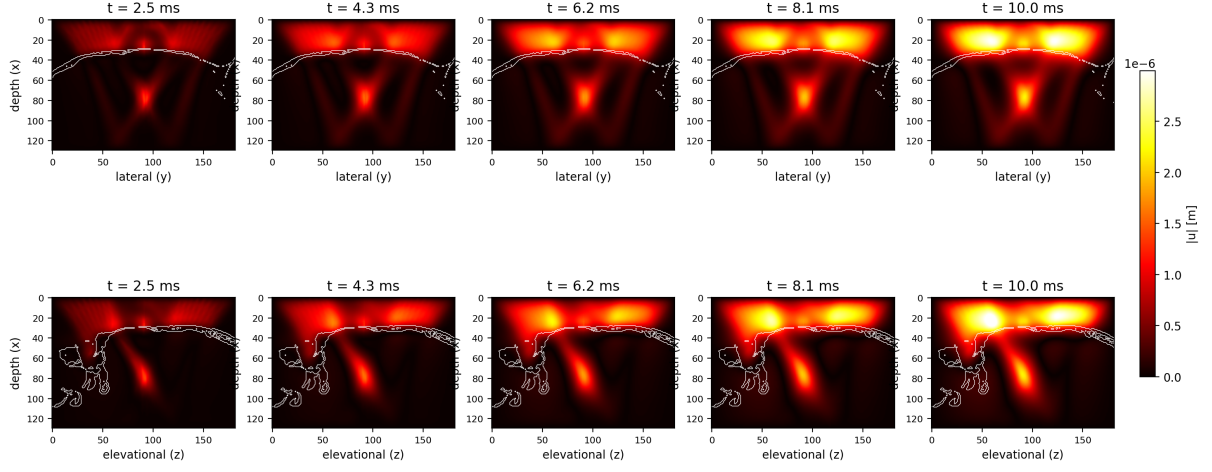


Figure 12: Shear wave displacement magnitude snapshots. **Top row:** x - y slices at center elevation. **Bottom row:** x - z slices at center lateral position. The displacement field expands radially from the focal spot over 0.7–3.0 ms. White contours indicate the skull boundary.

Figure 13 provides a 3D rendering of the displacement magnitude, showing the spatial extent of the shear wave field.

(f) 3D Displacement Magnitude

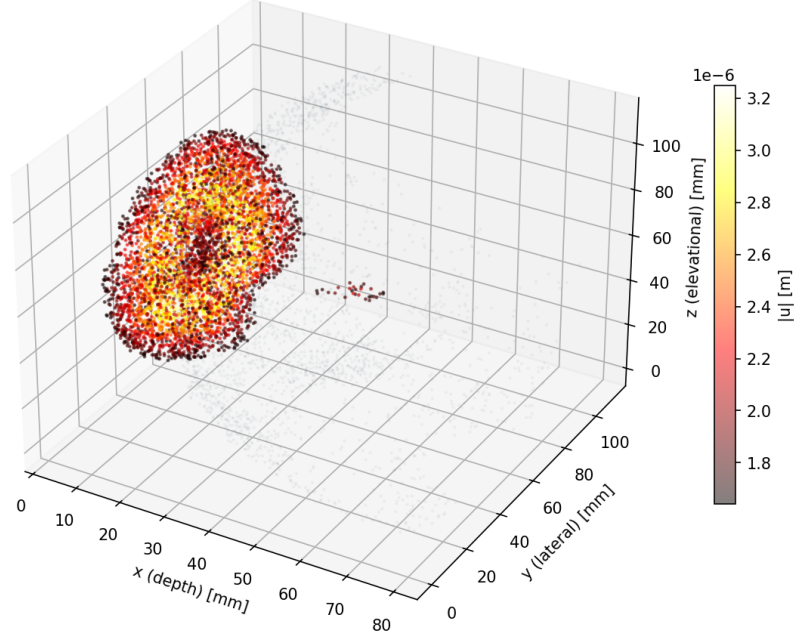


Figure 13: 3D rendering of the displacement magnitude field. The isosurface at 50% of peak $|u|$ is shown in coral, with the skull rendered as a translucent gray surface. The focal displacement region is clearly localized in the brain tissue volume behind the skull.

Figure 14 shows the temporal evolution of the displacement at the peak radiation force location, illustrating the pulsed tissue response to a single 2 ms push followed by 8 ms of viscous relaxation (20% duty cycle, 10 ms total). Peak displacement reaches $|u_x|^{\max} \approx 1.7 \mu\text{m}$.

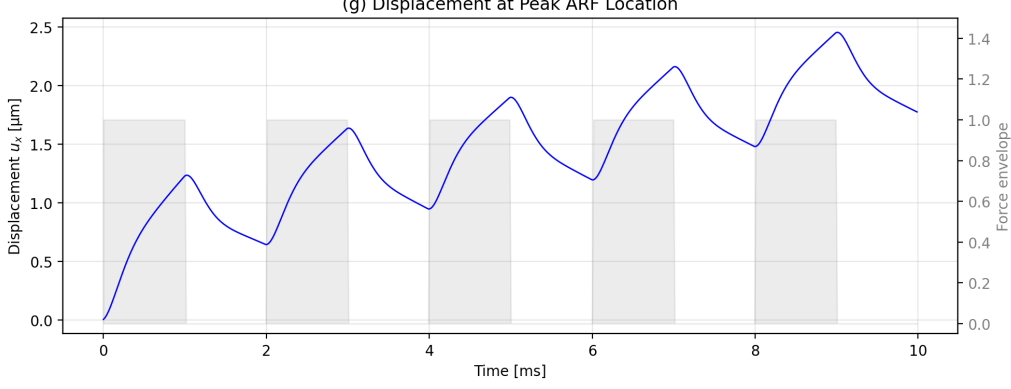


Figure 14: Displacement u_x at the peak ARF location as a function of time. The gray shading indicates the force envelope (single 2 ms push). The tissue displacement rises during the forcing interval and relaxes viscously during the subsequent 8 ms off period.

A.6 Theta-Burst TUS Protocol

The pipeline also supports a theta-burst transcranial ultrasound stimulation (TUS) protocol motivated by neuromodulation applications [17, 34]. Theta-burst TUS adapts the theta-burst pattern used in transcranial magnetic stimulation (TMS) to focused ultrasound, delivering brief acoustic pulses grouped at the 5 Hz theta rhythm that is prominent in hippocampal and cortical circuits.

Protocol parameters. The theta-burst envelope consists of $360 \mu\text{s}$ tone-bursts repeated at a 1 kHz pulse repetition frequency (PRF), grouped into trains of 3 bursts. Each 3-burst group occupies the first 3 ms of a 200 ms theta period (1/5 Hz), giving an overall duty cycle of $\sim 0.5\%$. One complete theta cycle (200 ms) is simulated.

Intensity scaling. The acoustic simulation described in Section A.6 was performed at a source pressure of 1.5 MPa, yielding a measured focal spatial-peak pulse-average intensity of $I_{\text{SPPA}} \approx 633 \text{ W/cm}^2$ (after CW envelope correction). Typical neuromodulation protocols operate at lower intensities, $I_{\text{SPPA}} \approx 20\text{--}30 \text{ W/cm}^2$ [17, 34]. Rather than re-running the acoustic simulation, the body force is scaled by the ratio of target-to-simulated focal intensity:

$$\mathbf{f}_{\text{neuro}} = \frac{I_{\text{SPPA}}^{\text{target}}}{I_{\text{SPPA}}^{\text{sim}}} \mathbf{f}_{\text{sim}}, \quad (51)$$

since the radiation force is proportional to acoustic intensity. For a target of 30 W/cm^2 the scaling factor is $30/633 \approx 0.047$, reducing the peak axial force from $\sim 41,800$ to $\sim 2,000 \text{ N/m}^3$.

Results. With the scaled ARF and theta-burst envelope, the shear FDTD solver (12035 time steps over 200 ms) produces a peak displacement of $|u_x|^{\text{max}} \approx 0.3 \text{ nm}$ at the end of the theta cycle. Instantaneous displacement during the 3 ms burst window reaches $\sim 0.08 \mu\text{m}$ before viscous damping ($\eta = 0.5 \text{ Pa}\cdot\text{s}$) dissipates the motion during the subsequent 197 ms off-period. The sub-nanometer residual at cycle end is consistent with the very low duty cycle (0.5%) and confirms that each theta cycle effectively returns the tissue to rest before the next group arrives.

Physical interpretation. The displacement magnitudes are orders of magnitude below those produced by clinical ARFI pushes (typically $1\text{--}20 \mu\text{m}$), which is expected: neuromodulation operates at much lower intensities and duty cycles. The neuromodulatory mechanism is not

believed to depend on large tissue displacements but rather on the mechanical perturbation of ion channels and membrane dynamics at the cellular scale [7, 30]. The theta-burst temporal structure provides a means to entrain neural oscillatory activity at the theta frequency while keeping time-averaged intensity low for safety.

A.7 Summary

This demonstration validates the complete sequential pipeline—acoustic transmit, radiation force computation, and shear wave FDTD—in a realistic transcranial geometry. The combined pulse-length and CW envelope normalization (Eqs. 48–49) provides a consistent mapping from the transient acoustic simulation to equivalent continuous-wave intensity, shared by both the radiation force and bioheat modules. The velocity-based Poynting vector ARF formulation (Eq. 50) captures off-axis energy flux near the focal zone, yielding significant lateral and elevational force components ($\sim 25\%$ of the peak axial force). A single 2 ms push produces a peak displacement of $\sim 1.7 \mu\text{m}$, with the shear wave expanding radially from the focal zone at the theoretical shear wave speed.

The theta-burst TUS mode (Section A.6) extends the pipeline to neuromodulation applications by scaling the body force to a target focal I_{SPPA} and applying a physiologically motivated temporal envelope (360 μs bursts at 1 kHz PRF, grouped at 5 Hz theta).

The pipeline is implemented in a single script with command-line options for skipping previously computed stages (`--skip-acoustic`, `--skip-shear`), selecting the ARF formulation (`--use-velocity`), and enabling the theta-burst protocol (`--theta-burst`), facilitating iterative parameter studies.

B TIPS Transcranial Focused Ultrasound Demo

This appendix presents a second end-to-end demonstration using the Philips TIPS (Therapeutic Imaging and Phased array Sonication) transducer, a curved-bowl annular phased array designed for transcranial focused ultrasound therapy and neuromodulation. The TIPS geometry differs fundamentally from the sparse phased array of Appendix A: rather than discrete point-like elements on a flat aperture, the TIPS uses a spherical-cap bowl surface with concentric annular ring elements, producing a geometrically focused beam that is then electronically steered via per-element delays.

The pipeline follows the same four-stage sequential workflow:

1. TIPS curved-bowl acoustic transmit through micro-CT skull (GPU FDTD),
2. Computation of acoustic radiation force from the pressure field,
3. Shear wave FDTD simulation of viscoelastic brain tissue displacement,
4. Visualization of all intermediate and final fields.

The corresponding script is `examples/wave_3d/tips_transcranial_shear_3d.py`.

B.1 TIPS Transducer Geometry

The TIPS transducer is modeled as a spherical-cap bowl with the following parameters: radius of curvature $R = 80 \text{ mm}$, inner aperture radius $r_{\text{in}} = 20.5 \text{ mm}$ (hollow center), outer aperture radius $r_{\text{out}} = 46 \text{ mm}$ (92 mm full diameter), 8 concentric annular elements of equal radial width, and a bowl thickness of 3 voxel layers along the surface normal. Per-pixel geometric focusing delays steer the beam to a 50 mm depth focal point, with delays averaged within each annular element to approximate the hardware phasing of the physical transducer.

Figure 15 shows the TIPS bowl source mask in the central elevational slice, illustrating the concentric annular geometry.

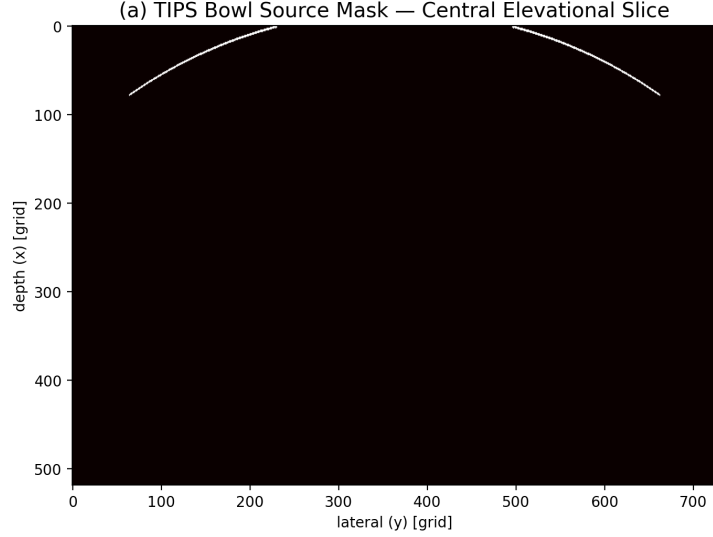


Figure 15: TIPS bowl source mask in the x - y plane (central elevational slice). The concentric annular ring structure and hollow center are clearly visible. The bowl curvature produces a crescent cross-section in the depth-lateral plane.

B.2 Acoustical Medium and Pressure Propagation

The heterogeneous skull medium uses the same Halle temporal bone micro-CT specimen and `MicroCTSkullDomain` preset as Appendix A. Because the TIPS bowl surface extends further in depth than a flat aperture, the skull slab is shifted deeper to avoid geometric intersection between the transducer source voxels and the bone.

The acoustic simulation uses a source pressure amplitude of 400 kPa (lower than the 1.5 MPa used for the sparse array, reflecting neuromodulation operating levels) with 3-cycle Gaussian-modulated sinusoidal pulses at $f_0 = 1$ MHz. The domain size is $80 \times 112 \times 112$ mm with 10 points per wavelength.

Figure 16 shows the acoustical medium property maps, and Figure 17 shows the focused beam propagating through the skull.

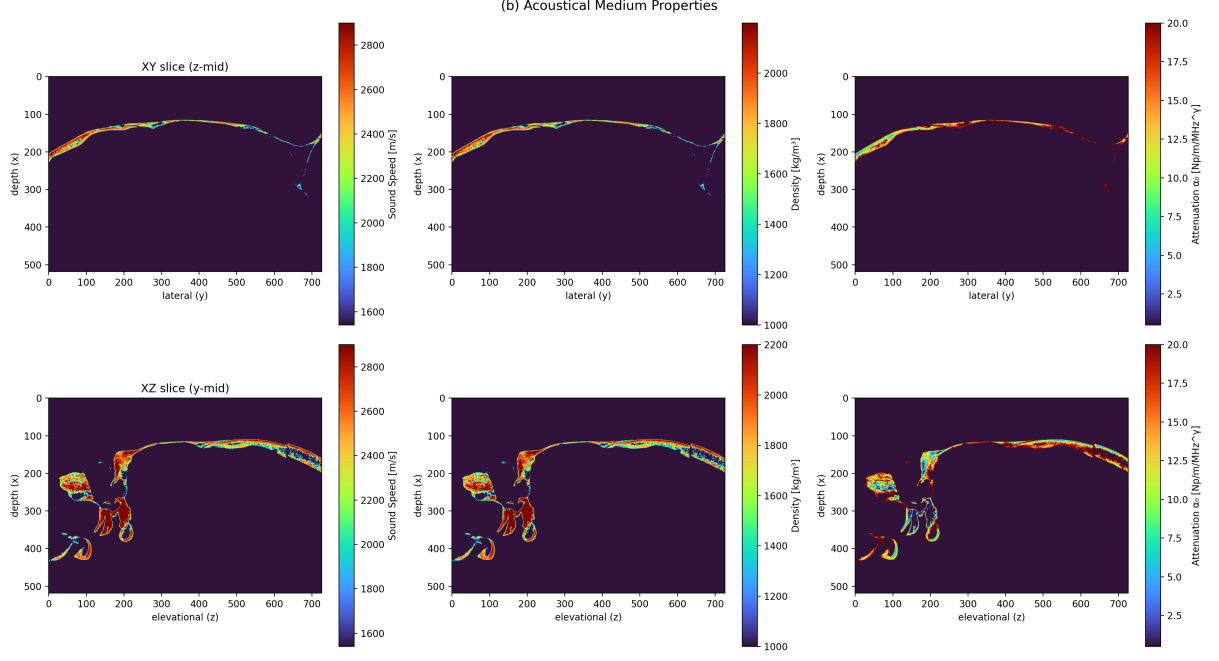


Figure 16: Acoustical medium properties for the TIPS transcranial configuration. **Top row:** x – y slices at center elevation. **Bottom row:** x – z slices at center lateral position. The skull layer appears as the high-speed, high-density region, shifted deeper relative to the sparse array configuration to clear the TIPS bowl geometry.

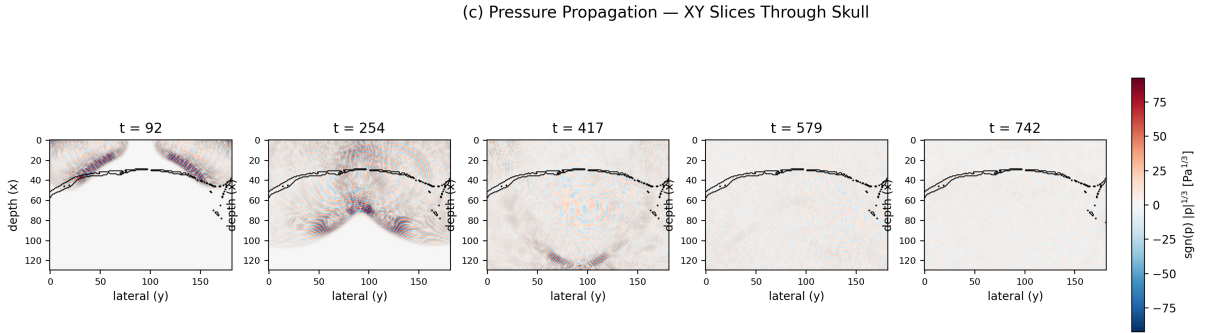


Figure 17: Pressure propagation snapshots for the TIPS focused transmit. The geometrically focused bowl produces a converging wavefront that propagates through the skull and focuses at 50 mm depth. Compared to the sparse array (Figure 10), the TIPS bowl produces a smoother wavefront due to its continuous annular aperture.

B.3 Acoustic Radiation Force

The acoustic radiation force is computed using the plane-wave approximation with the same pulse-length and CW envelope normalization (Eqs. 48–49) as the sparse array demo. Radiation force within the skull is masked to zero. The peak body force after skull masking is $b_0^{\max} \approx 22,600 \text{ N/m}^3$. Figure 18 shows the resulting body force distribution.

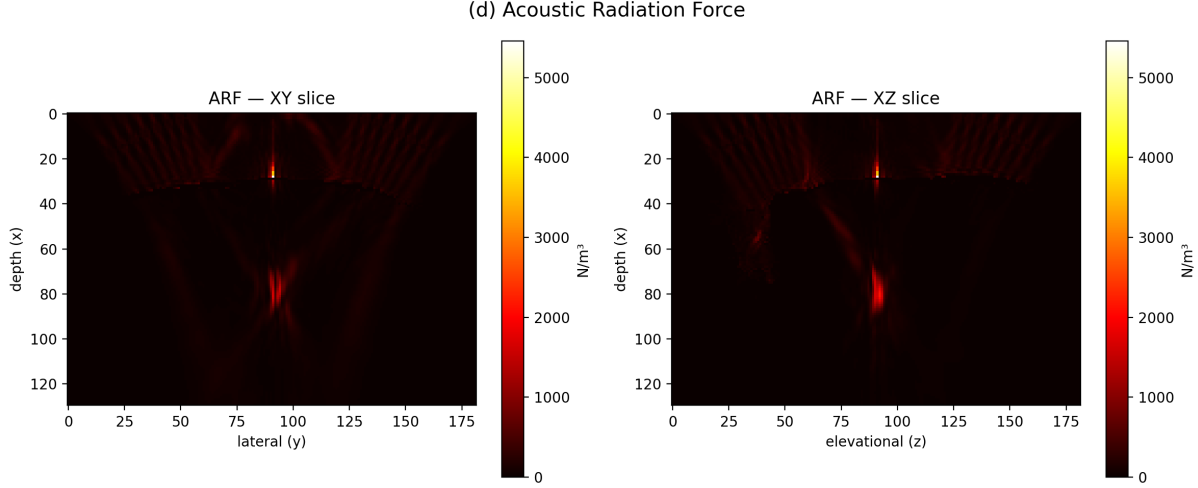


Figure 18: Acoustic radiation force (depth component b_0) for the TIPS focused transmit. **Left:** x - y slice. **Right:** x - z slice. The force is concentrated at the 50 mm focal spot, with the skull region masked to zero.

B.4 Shear Wave Displacement

The shear wave simulation uses the same brain tissue parameters as the sparse array demo (Appendix A). The pulsed forcing protocol uses a 50% duty cycle (1 ms on, 1 ms off, 5 pulses over 10 ms total), modulated by the scalar temporal envelope $g(t)$ via the pre-computed Helmholtz projection (Section 3.3).

Peak displacement reaches $|u_x|^{\max} \approx 3.3 \mu\text{m}$. Figure 19 shows the displacement magnitude field at multiple time points, demonstrating shear wave expansion from the TIPS focal zone.

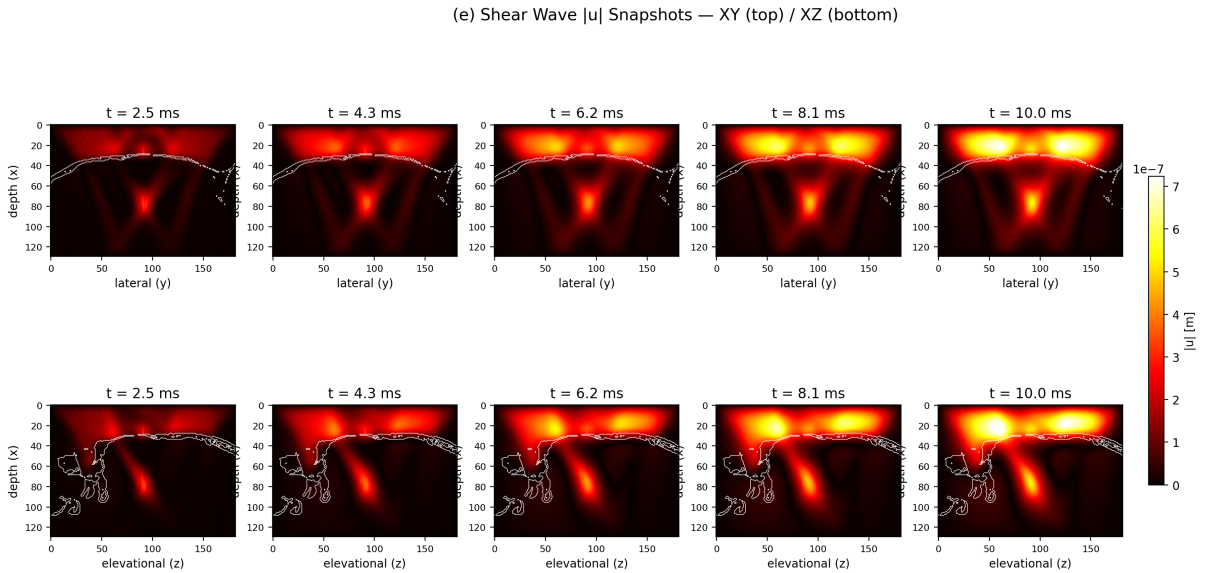


Figure 19: Shear wave displacement magnitude snapshots for the TIPS configuration. **Top row:** x - y slices at center elevation. **Bottom row:** x - z slices at center lateral position. White contours indicate the skull boundary. The displacement pattern reflects the TIPS focal geometry.

Figure 20 provides a 3D rendering of the displacement field.

(f) 3D Displacement $|u| = 0.4 \mu\text{m}$

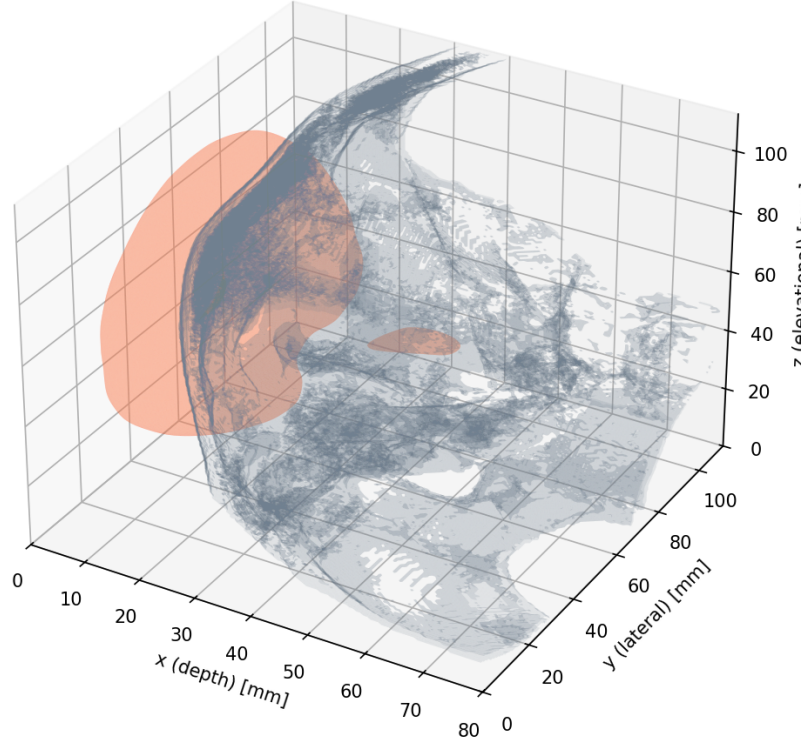


Figure 20: 3D rendering of the TIPS displacement magnitude. The displacement region is localized in brain tissue behind the skull, with the skull rendered as a translucent gray surface.

Figure 21 shows the temporal displacement trace at the peak ARF location, illustrating the pulsed tissue response.

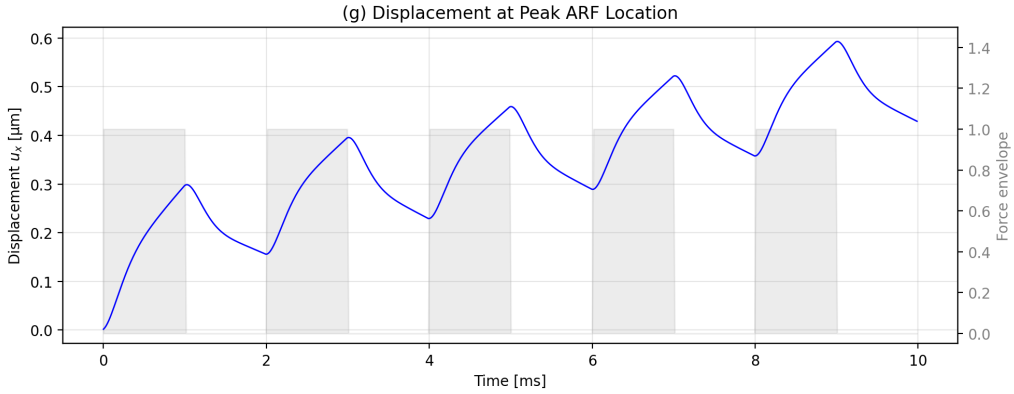


Figure 21: Displacement u_x at the peak ARF location for the TIPS configuration. The gray shading indicates the pulsed force envelope (50% duty cycle, 1 ms on / 1 ms off). Each pulse drives tissue displacement followed by viscous relaxation during the off period.

B.5 Comparison with Sparse Phased Array

The TIPS and sparse array configurations represent two distinct approaches to transcranial focused ultrasound. The key differences relevant to shear wave generation are:

- **Aperture geometry.** The TIPS bowl (92 mm diameter, spherical cap, 8 annular elements) produces geometric focusing via the curved surface, augmented by electronic delays. The sparse array (1024 point elements on a flat aperture) relies entirely on electronic focusing.
- **Source pressure.** The TIPS demonstration uses 400 kPa source pressure, compared to 1.5 MPa for the sparse array, reflecting different operating regimes (neuromodulation vs. clinical ARFI).
- **Focal quality.** The continuous annular aperture of the TIPS produces smoother wavefronts through the skull compared to the sparse array, which can exhibit grating-lobe-like artifacts from the discrete element distribution.
- **Skull interaction.** The TIPS bowl extends further in depth, requiring the skull slab to be shifted to avoid geometric intersection. This changes the effective skull thickness and angle of incidence relative to the sparse array configuration.

Despite these differences, the sequential pipeline (acoustic FDTD $\rightarrow$ ARF $\rightarrow$ shear FDTD) and the shear wave physics remain identical, demonstrating the generality of the reduced elastodynamic framework for different transducer configurations.

B.6 Thermal Neuromodulation Coupling

The TIPS pipeline is further extended with bioheat-coupled mechanosensitive neuron simulations, using the `fullwave.neuromodulation` module to predict neural firing from the shear wave displacement field. The heat source is computed using the same pulse-length and CW envelope normalization (Eqs. 48–49) and scaled by the duty cycle. A cortical pyramidal neuron model (Pospischil et al., 2008) with Piezo1 and TREK-1 mechanosensitive channels predicts spatially selective firing in brain tissue at the mechanical focus. Full details of the neuron model, channel parameterization, and firing results are presented in the companion neuromodulation report.

B.7 Acoustic Strain and Strain Gradient

The preceding sections compute the *elastic* (shear-wave) strain produced by the slow radiation-force-driven tissue displacement (ms timescale, $\sim 1\text{--}5$ m/s shear speed). A distinct and complementary quantity is the *acoustic* strain produced directly by the MHz pressure wave itself. From linearized acoustics, the volumetric acoustic strain at any point and instant is related to the local pressure by the equation of state

$$\varepsilon_{ac}(\mathbf{x}, t) = -\frac{p(\mathbf{x}, t)}{K(\mathbf{x})}, \quad K = \rho c^2, \quad (52)$$

where K is the bulk modulus. For soft tissue ($\rho \approx 1000$ kg/m³, $c \approx 1540$ m/s), $K \approx 2.4$ GPa. Because K is six orders of magnitude larger than the shear modulus ($\mu \approx 5$ kPa), the acoustic strain is extremely small even at therapeutic pressures. This relation is instantaneous and exact in the linear regime—no time averaging or pulse-length normalization is required.

We compute two summary fields from the TIPS pressure output:

- **Peak acoustic strain:** $\varepsilon_{ac}^{\max}(\mathbf{x}) = \max_t |p(\mathbf{x}, t)|/K(\mathbf{x})$,
- **Peak strain gradient magnitude:** $|\nabla \varepsilon_{ac}|^{\max}(\mathbf{x}) = \max_t |\nabla \varepsilon_{ac}(\mathbf{x}, t)|$,

where the spatial gradient $\nabla \varepsilon_{ac}$ is computed via central differences on the downsampled grid. The gradient is taken of the full strain field $\varepsilon_{ac} = -p/K$ (rather than just the pressure gradient) so that spatial variations of the bulk modulus at tissue interfaces are captured. To suppress

finite-difference artifacts at the skull–tissue boundary, the skull mask is dilated by one voxel before zeroing strain and gradient values inside bone.

For the TIPS configuration with 400 kPa source pressure at 1 MHz through the skull, we obtain:

- Peak acoustic strain: $\varepsilon_{ac}^{\max} \approx 8.8 \times 10^{-4}$
- Peak strain gradient: $|\nabla \varepsilon_{ac}|^{\max} \approx 1.5 \text{ m}^{-1}$

Figure 22 shows the peak acoustic strain distribution. The strain field mirrors the focal pressure pattern, with the highest values at the geometric focus and along the beam axis within the skull aperture.

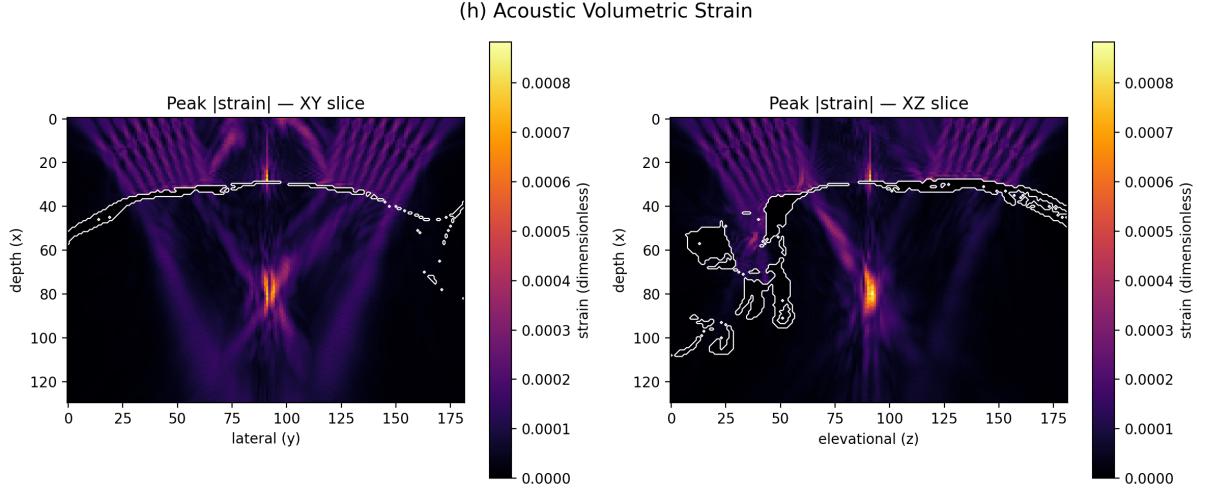


Figure 22: Peak volumetric acoustic strain $\varepsilon_{ac}^{\max} = \max_t |p|/K$ for the TIPS configuration. **Left:** x – y slice at center elevation. **Right:** x – z slice at center lateral position. White contours indicate the skull boundary. The strain pattern reflects the acoustic focal geometry.

Figure 23 shows the peak strain gradient magnitude. The gradient is largest at the boundaries of the focal zone and at tissue–bone interfaces where bulk modulus changes abruptly, rather than at the pressure maximum itself.

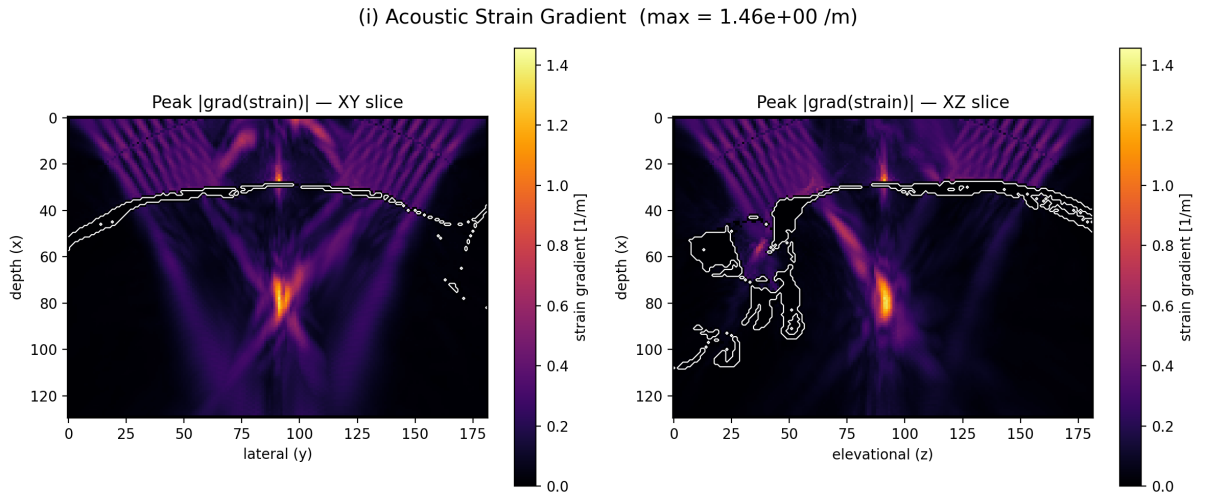


Figure 23: Peak acoustic strain gradient magnitude $|\nabla \varepsilon_{ac}|^{\max}$ for the TIPS configuration. **Left:** x – y slice. **Right:** x – z slice. The gradient is concentrated at the edges of the focal zone rather than at the pressure peak.

B.8 Acoustic vs. Elastic Strain: Comparison and Neuromodulation Mechanisms

Table 2 compares the acoustic (pressure-wave) and elastic (shear-wave) strain fields computed from the TIPS transcranial simulation.

Table 2: Comparison of acoustic and elastic strain magnitudes for the TIPS transcranial configuration (400 kPa source, 1 MHz, skull-masked).

Quantity	Acoustic (pressure wave)	Elastic (shear wave)
Peak volumetric strain	8.8×10^{-4}	1.8×10^{-4}
Peak deviatoric strain	—	4.4×10^{-4}
Peak von Mises strain	—	3.6×10^{-4}
Peak max shear strain	—	3.1×10^{-4}

The two strain fields are of the same order of magnitude ($\sim 10^{-4}$), with the peak acoustic volumetric strain roughly $5\times$ larger than the elastic volumetric strain and $2.5\times$ larger than the elastic von Mises strain. Despite this quantitative similarity, the two fields differ fundamentally in their spatiotemporal character and the neuromodulation mechanisms they support.

B.8.1 Timescale separation

The *acoustic strain* oscillates at the ultrasound carrier frequency ($f_0 = 1$ MHz, period $1\ \mu\text{s}$). Each acoustic cycle produces a compressive–rarefactive strain cycle that acts on the membrane for sub-microsecond durations. The *elastic strain* evolves on the shear wave timescale—the shear speed in brain tissue is $c_s \approx 2$ m/s, so shear wavelengths at the focal zone scale (~ 5 mm) correspond to periods of ~ 2.5 ms. The pulsed forcing protocol (1 ms on, 1 ms off) further shapes the temporal envelope of the elastic strain. The ratio of timescales is $\sim 10^3$, matching the ratio of longitudinal to shear wave speeds.

B.8.2 Mechanosensitive ion channels (elastic strain mechanism)

The shear-wave displacement field acts through the elastic strain tensor to produce membrane tension (Eq. 47). Mechanosensitive channels such as Piezo1 ($T_{1/2} \approx 2.7$ mN/m) and TREK-1 ($T_{1/2} \approx 3.5$ mN/m) gate on the ms timescale, consistent with the shear wave period. The deviatoric strain ($\sim 4 \times 10^{-4}$) acts through the area-expansion modulus K_A to produce membrane tension. For $K_A = 0.25$ N/m, this yields $T \approx K_A \varepsilon_{\text{dev}} \approx 0.11$ mN/m—below the half-activation tensions but in the range where cumulative or resonant effects may contribute over multiple pulsed cycles [31]. This mechanism is intrinsically *mechanical*: it requires the ARF-driven tissue displacement and is mediated by the shear modulus of the tissue.

B.8.3 Flexoelectric coupling (acoustic strain gradient mechanism)

A distinct hypothesis, developed by Felix et al. [21], proposes that *acoustic strain gradients* directly polarize neural membranes through the flexoelectric effect. In this model, the spatial gradient of the acoustic volumetric strain $\nabla \varepsilon_{\text{ac}}$ produces a polarization density in the lipid bilayer proportional to the flexoelectric coefficient μ_f :

$$P = \mu_f \nabla \varepsilon_{\text{ac}}, \quad (53)$$

where μ_f has units of C/m and typical values for lipid membranes are estimated at $\sim 10^{-18}$ – 10^{-17} C/m [21]. This polarization acts as an effective transmembrane current that can depolarize or hyperpolarize the membrane, potentially modulating neural firing without requiring any tissue displacement or mechanosensitive channel activation.

Several features distinguish this mechanism:

- **Timescale:** The flexoelectric effect operates at the acoustic frequency (MHz), orders of magnitude faster than ion channel kinetics. The effective current is the *time-averaged* rectified polarization, whose amplitude depends on the strain gradient envelope rather than individual acoustic cycles.
- **Spatial pattern:** The strain gradient is largest at the *edges* of the focal zone (where the pressure field changes most rapidly in space), not at the pressure peak. This is visible in Figure 23 and contrasts with the ARF, which peaks at the pressure maximum.
- **Independence from tissue stiffness:** The acoustic strain depends on the bulk modulus $K = \rho c^2$ (GPa scale), not the shear modulus μ (kPa scale). The effect is therefore insensitive to variations in tissue viscoelastic properties that strongly modulate the shear-wave response.
- **No displacement required:** Because the mechanism operates directly on the propagating acoustic wave, it does not require the intermediate step of ARF-driven tissue displacement. Any pressure wave—including reflected and scattered components—contributes to the strain gradient field.

B.8.4 Comparison figure

Figure 24 provides a side-by-side comparison of the four spatial fields relevant to neuromodulation:

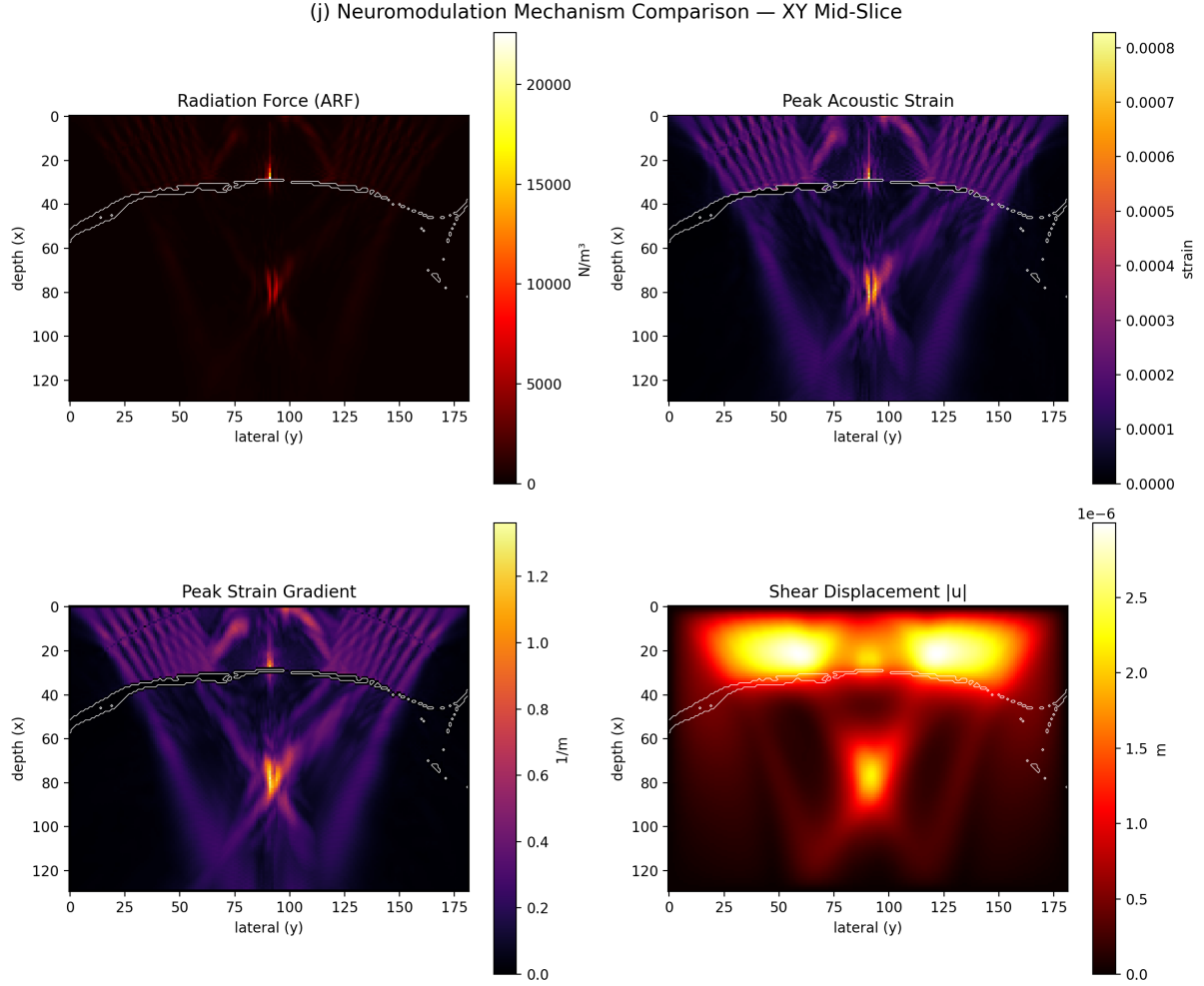


Figure 24: Comparison of neuromodulation-relevant fields in the x - y mid-slice. **Top left:** acoustic radiation force (ARF), the driving source for shear wave generation. **Top right:** peak acoustic strain, directly proportional to peak pressure. **Bottom left:** peak acoustic strain gradient magnitude, relevant for flexoelectric coupling. **Bottom right:** shear wave displacement magnitude, the end result of the ARF→shear wave pipeline. White contours indicate the skull boundary. Note the qualitatively different spatial patterns: ARF and acoustic strain peak at the focal center, while the strain gradient is largest at the focal boundary; shear displacement spreads over a broader region due to wave propagation.

The four fields have qualitatively different spatial patterns:

1. **ARF and acoustic strain** peak at the geometric focus where pressure is highest, producing compact, symmetric distributions.
2. **Strain gradient** peaks at the boundary of the focal zone where the pressure spatial derivative is largest, forming a ring-like or shell-like pattern around the focus.
3. **Shear displacement** spreads over a broader region than the ARF source due to elastic wave propagation, with the pattern shaped by the cumulative history of pulsed forcing and viscous relaxation.

These distinct spatial signatures suggest that the two neuromodulation mechanisms—mechanical (ARF → shear strain → membrane tension → channel gating) and electrical (acoustic strain gradient → flexoelectric polarization → transmembrane current)—would produce different spatial selectivity profiles. Experimental measurements of neural activation patterns during TUS

could in principle discriminate between these mechanisms, since the flexoelectric model predicts peak activation at the *edges* of the focal zone rather than at the center.

B.9 Summary

The TIPS transcranial demo validates the acoustic–elastic–neural pipeline with a clinically relevant curved-bowl transducer geometry. The combined pulse-length and CW envelope normalization (Eqs. 48–49) provides a consistent mapping from the transient simulation to CW-equivalent fields for both radiation force and bioheat. Peak body force after skull masking is $\sim 22,600 \text{ N/m}^3$ and peak shear wave displacement is $\sim 3.3 \text{ }\mu\text{m}$.

The acoustic strain analysis (Section B.7) reveals that the pressure-wave strain ($\varepsilon_{\text{ac}}^{\text{max}} \approx 8.8 \times 10^{-4}$) and the shear-wave elastic strain ($\varepsilon_{\text{dev}}^{\text{max}} \approx 4.4 \times 10^{-4}$) are of the same order of magnitude but differ by three orders of magnitude in timescale (MHz vs. ms). The strain gradient field ($|\nabla \varepsilon_{\text{ac}}|^{\text{max}} \approx 1.5 \text{ m}^{-1}$) exhibits a spatially distinct pattern from the ARF, peaking at the focal boundary rather than the center. These results enable quantitative comparison of the mechanical (ARF–shear–channel gating) and electrical (flexoelectric–strain gradient) hypotheses for ultrasonic neuromodulation.

The thermal neuromodulation extension couples the displacement field with bioheat-driven temperature rise and mechanosensitive neuron models, predicting spatially selective firing volumes localized at the acoustic focus.